

The Evolution of AI at Booking.com

Architectures You've Always Wondered About

QCon London 2026

INTRO

2005

EXPERIMENTATION

DATA

MYSQL

HADOOP

CLOUD

ML ENGINEERING

ML INFERENCE

FEATURE ENGINEERING

DOMAIN INTELLIGENCE

RANKING

SUMMARY

Navigation Tips

Many slides have **transitions/fragments** that reveal content step-by-step

To skip transitions: Use **Shift** + **←** / **→**
Normal navigation: Use **←** / **→** arrow keys

Today, Booking.com impacts

Billions

of travel decisions every day

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Today's Roadmap: **The Three Layers**

Domain Intelligence

ML Engineering

Data Management

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LET'S GO BACK TO 2005

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The World in 2005



Motorola Razr V3

The coolest phone you could own



Web 2.0 Era

User-generated content takes off

Booking.com

Booking.nl

9 years old

Inside Booking.com

First Official A/B Experiment

February 5, 2005

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Everything goes through experimentation

CSS tweaks



Infrastructure hanges

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1000+

Run In Parallel Anytime

150K+

Total Experiments

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Most of Our Experiments Fail

OUR SUCCESS RATE

<25%

The goal isn't to be right. It's to learn fast.

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Experimentation Built Our **Data-Driven DNA**

Isolated Code & Feature Releases • Minimized Risk
Eliminated HIPPOs • Ended Debates • Captured Learnings

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Let's begin here

Domain Intelligence

ML Engineering

Data Management



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Our Data Story

MySQL • Hadoop • Move to Cloud

The Original Tech Stack



Perl

+



MySQL

~ 2000

- ✓ Asynchronous replication
- ✓ Commercial support

MySQL Growth



What Is So Unique About Our MySQL Setup?

We **Don't** Have ...

- ✗ Specialized Hardware
- ✗ Stored Procedures
- ✗ UDFs
- ✗ Views
- ✗ Cache Layer

Wait... No Cache Layer?!

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The Secret Sauce

Smaller DBs

2TB soft limit

+

Fits in NVMe

of standard server blade 7/9

=

< 350 μ s

point queries

Tested on Blade 7 - 2TB

Faster backups · Quicker failovers · Lower cost

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How Do We Manage Thousands of DB Instances?

Single Writer

+

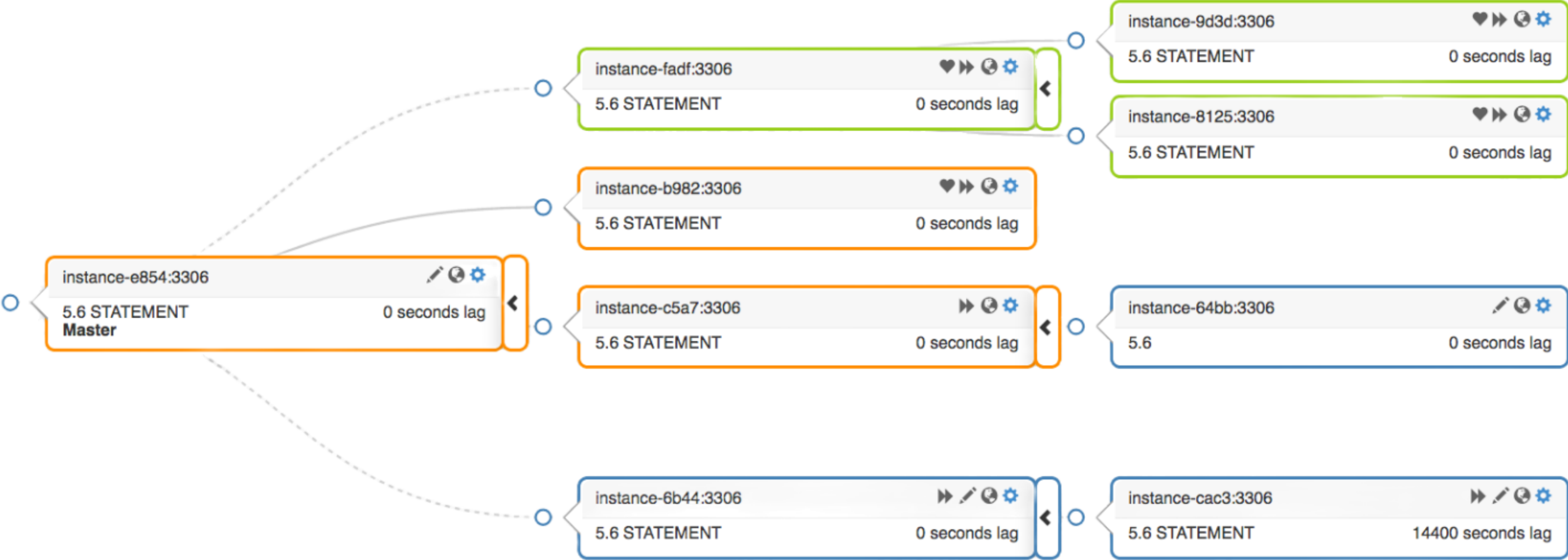
MySQL Orchestrator

+

Automation

Our MySQL Setup

Single writer · Replication chain · Automatic Failovers



The Need for Automation

Managing Refresh Cycles

$$\begin{array}{ccc} \text{INSTANCES} & & \text{REFRESH CYCLE} \\ \sim 7000 & \div & 60 \text{ days} \\ & = & \end{array} \quad \begin{array}{c} \text{RESULT} \\ \sim 1 \text{ action every 3 min} \end{array}$$

We didn't just **Use** MySQL — we helped
Shape it

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Early decision: **Don't fork,** **Contribute upstream**

Many modern MySQL features exist today because of us

GTID-based Replication

Binlog Compression (zstd)

Parallel Replication

...

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ML in Perl-Era



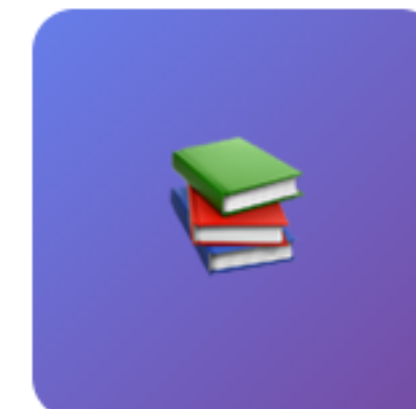
Perl Scripts

+



Cron Jobs

+



CPAN

Statistical libraries

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It solved real business problems

Classical ML Examples ~2011



Classification

Fraud Detection

Business vs Leisure

Review Categorization

Naive Bayes

Regex NLP



Prediction

Demand Forecasting

Curve-Fitting



Recommendations

Similar Hotels

Hotel-to-Hotel Recs

KNN + Fisher Test

Collaborative Filtering

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This Setup Worked, Until...

Data grew too large

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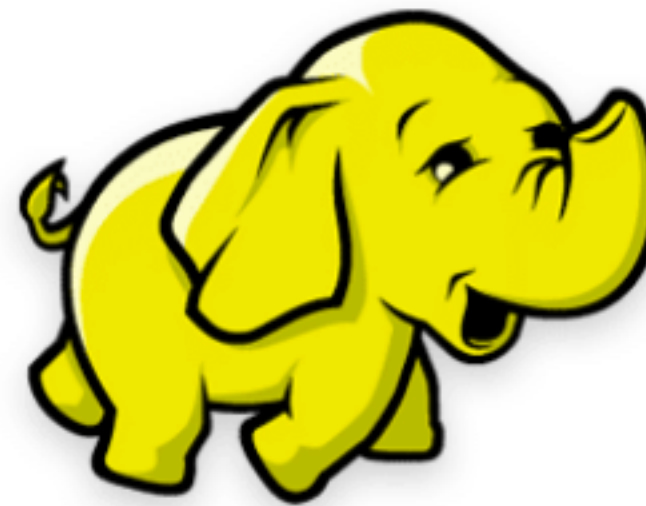
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Hadoop to the Rescue

Distributed storage and processing at scale

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2 x On-Premise Clusters (~2011)

Per-cluster configuration

CORES
~60,000

RAW DISK
200 PB

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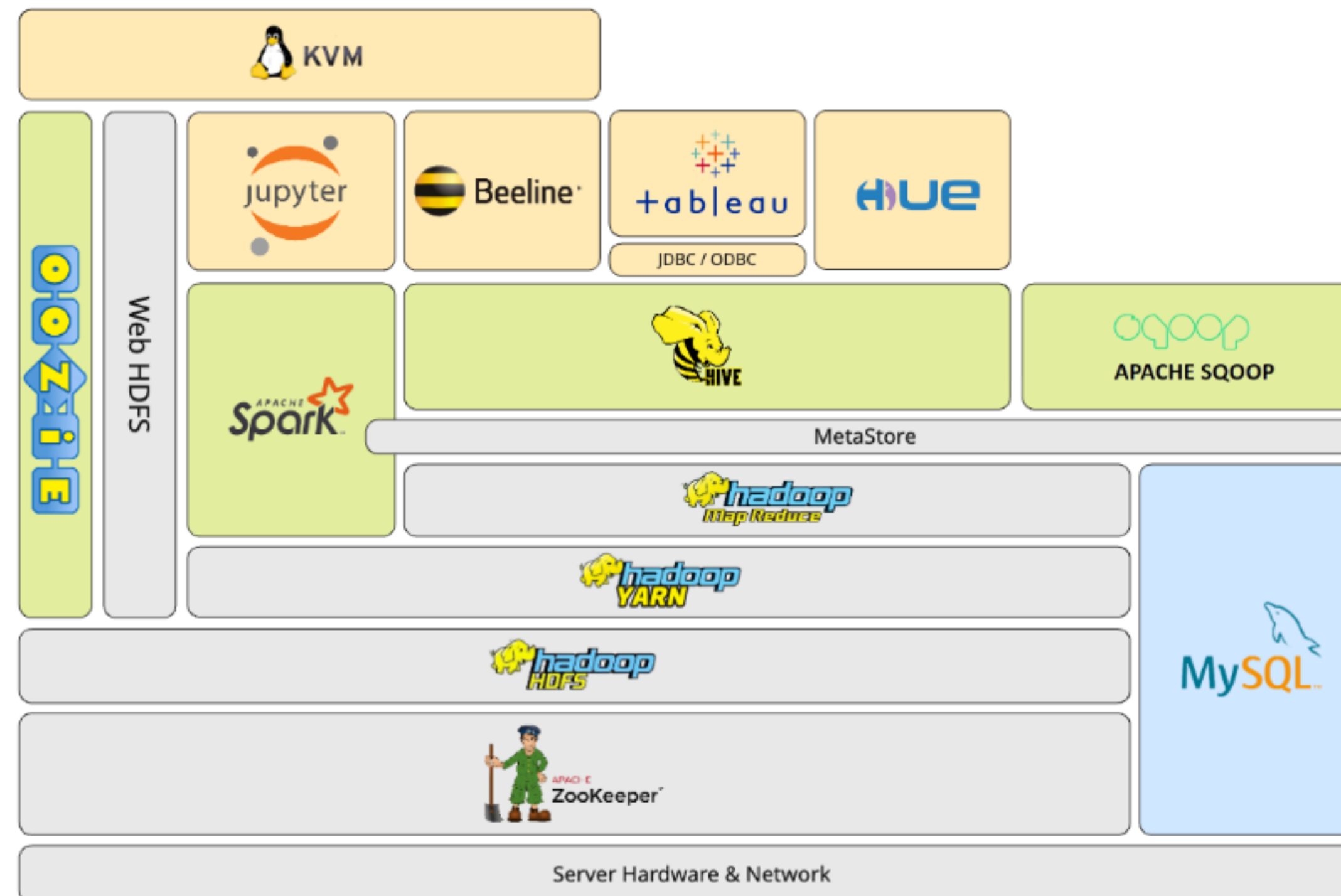
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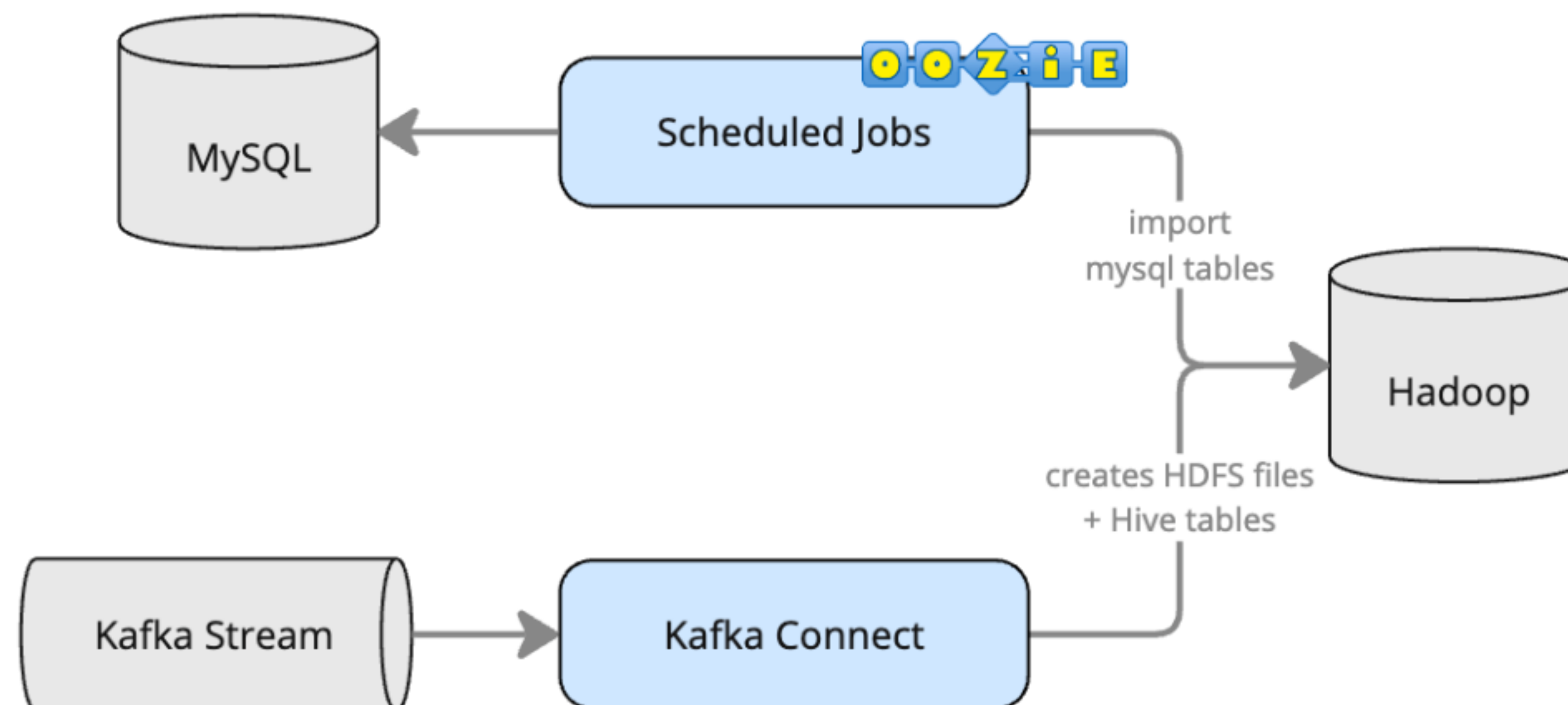
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The Setup Looked Like This

Our Hadoop ecosystem

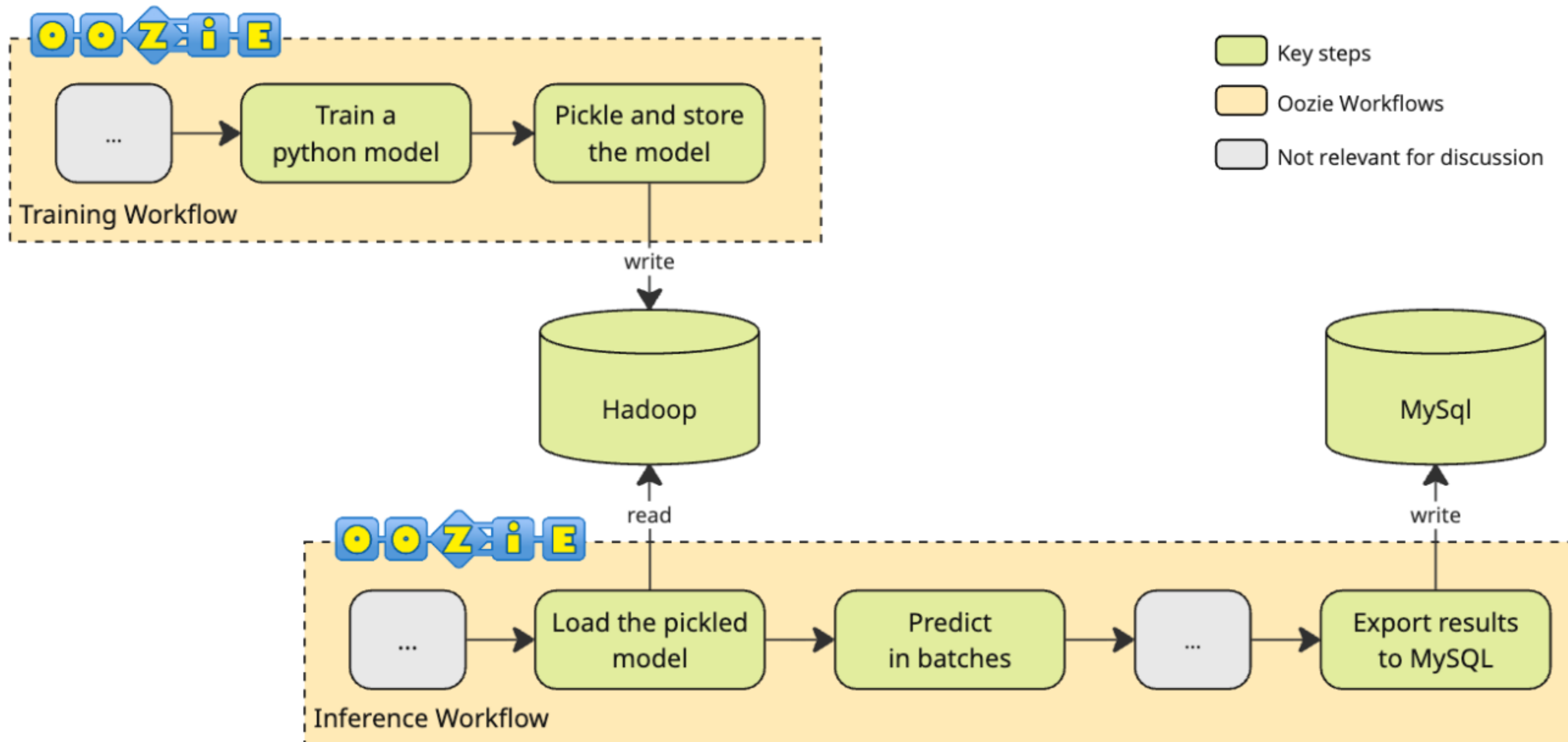


How Data Reaches Hadoop?

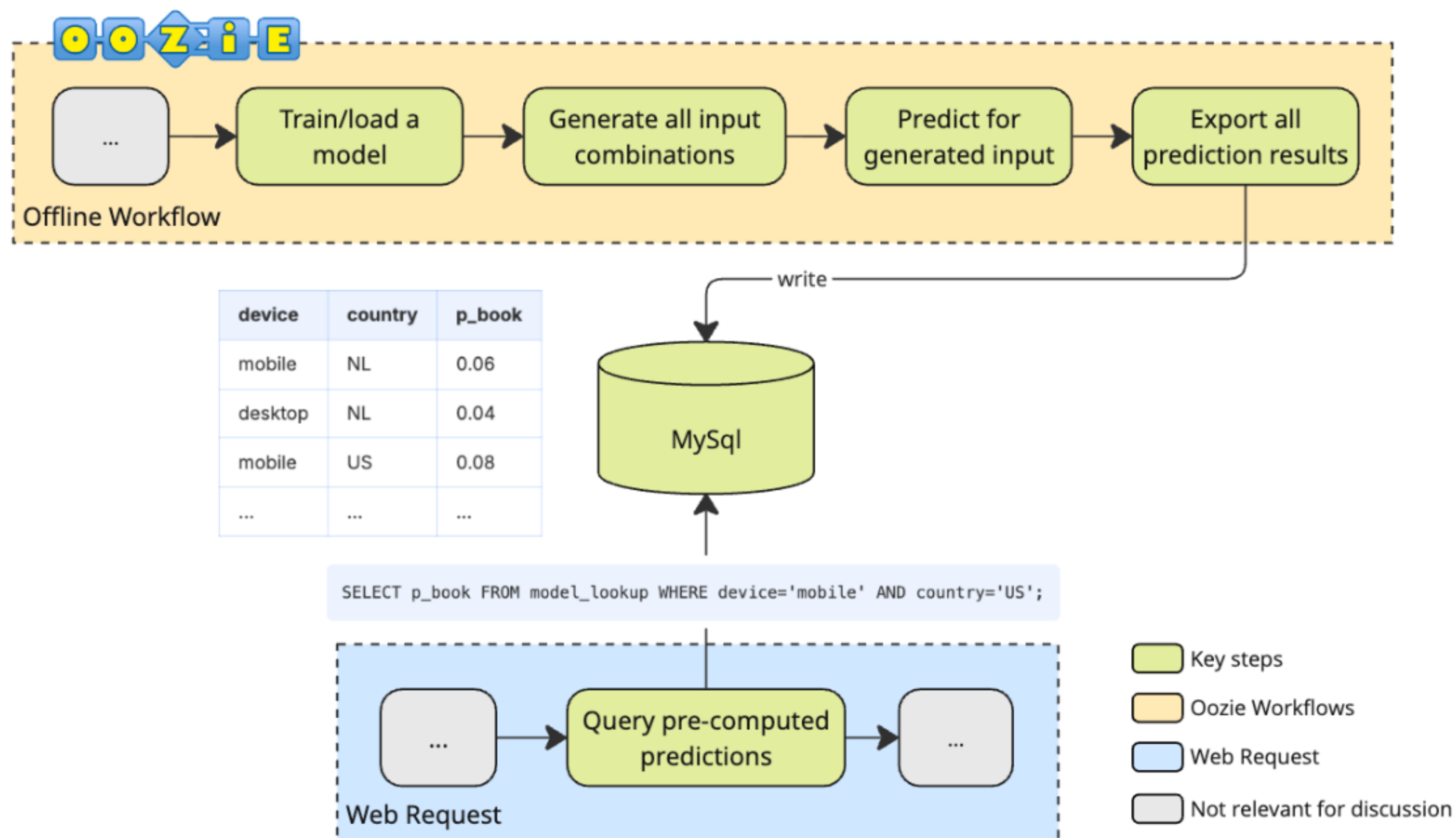


And This Data Powered ML Predictions

An example of offline inference



We faked "Realtime Predictions" with **Lookup Tables**



No real-time computation • MySQL as Cache

Hadoop Powered Our ML Pipeline For Years

But....

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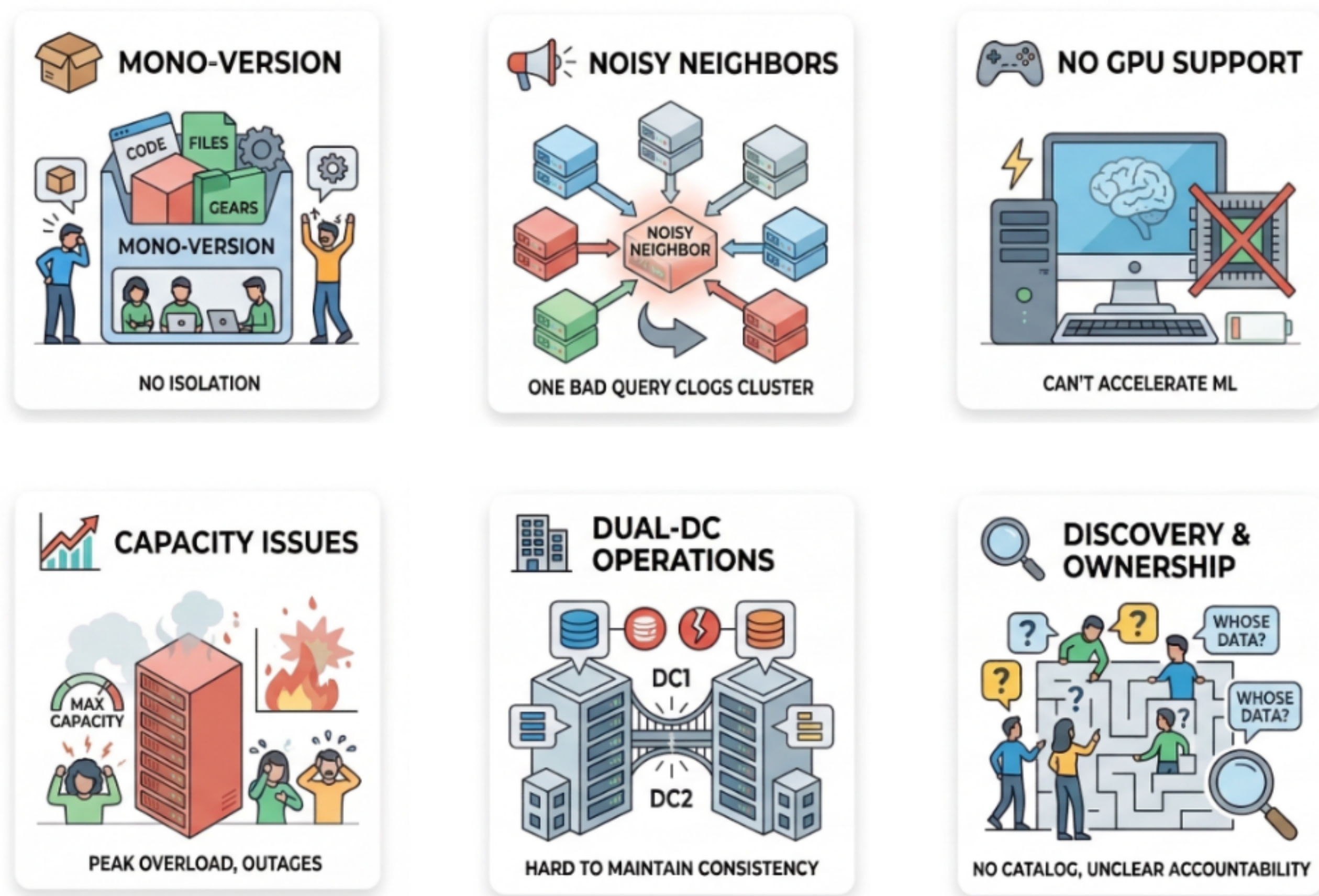
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... The Cracks Appeared

An ML scientist's perspective



Is this Mold?

Petabytes of data • Thousands of workflows • Circa 2018

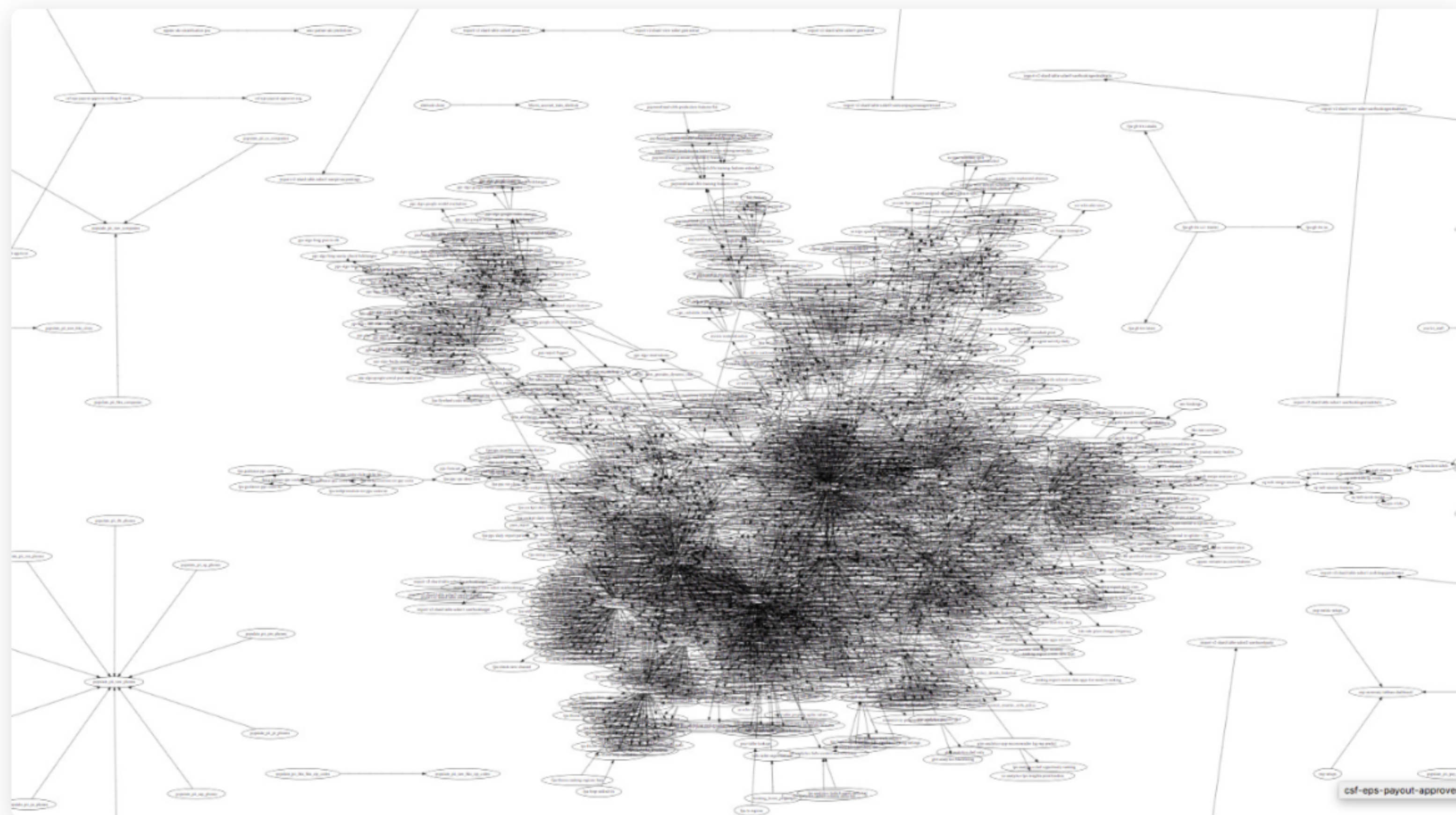
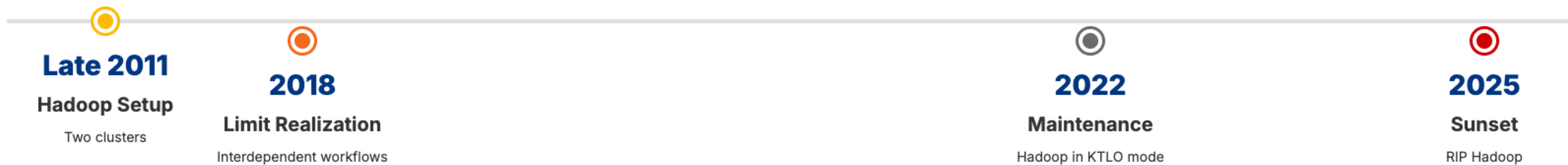


Image credit: Bennie Schut

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Saying Goodbye to Hadoop



** Indicative timeline*

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And The Odds Were Against Us



100s
of petabytes



~190K
Hadoop tables



7,000+
workflows



100s
of teams



24/7
continuity

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Our Goal

Move Everything • Stop Nothing • Align Everyone

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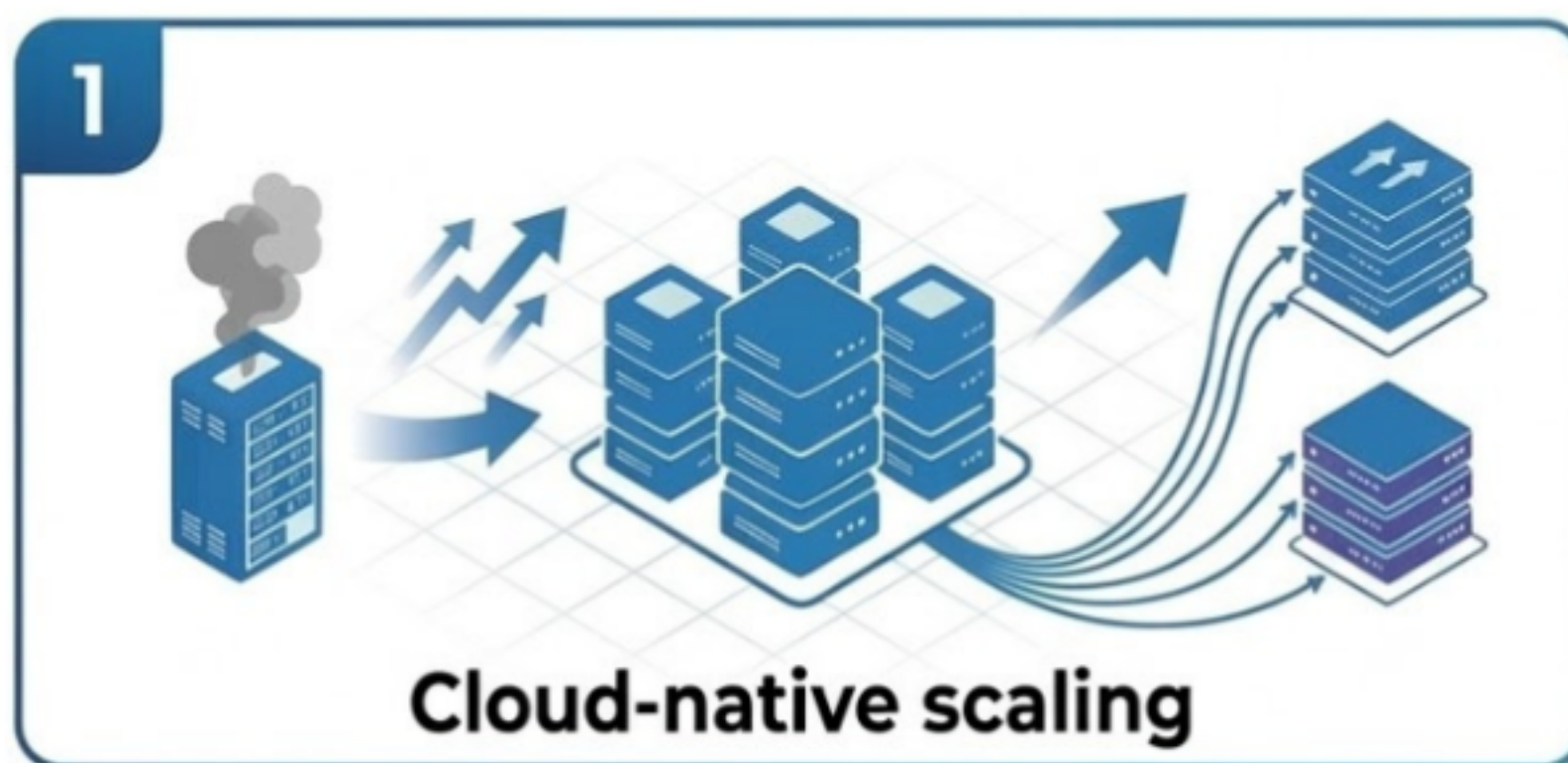
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The Four Major Hurdles



Saying Goodbye to Hadoop



* Indicative timeline



This is How it Happened

Five phases of migration

Mapped the Universe



Reduced Scope



Applied PageRank



Migrated in Waves



Phased Out Legacy





Key to Our Success: A Unified Command Center

Everything in one place — status, dependencies, blockers





To Pull-off the Impossible, **Burn the Boats**

When there is no plan B, we will find a way

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But a Better Strategy is **Prevention**

Four Lessons we'd hand our past selves



Catalogue Every Asset



Establish Clear Ownership

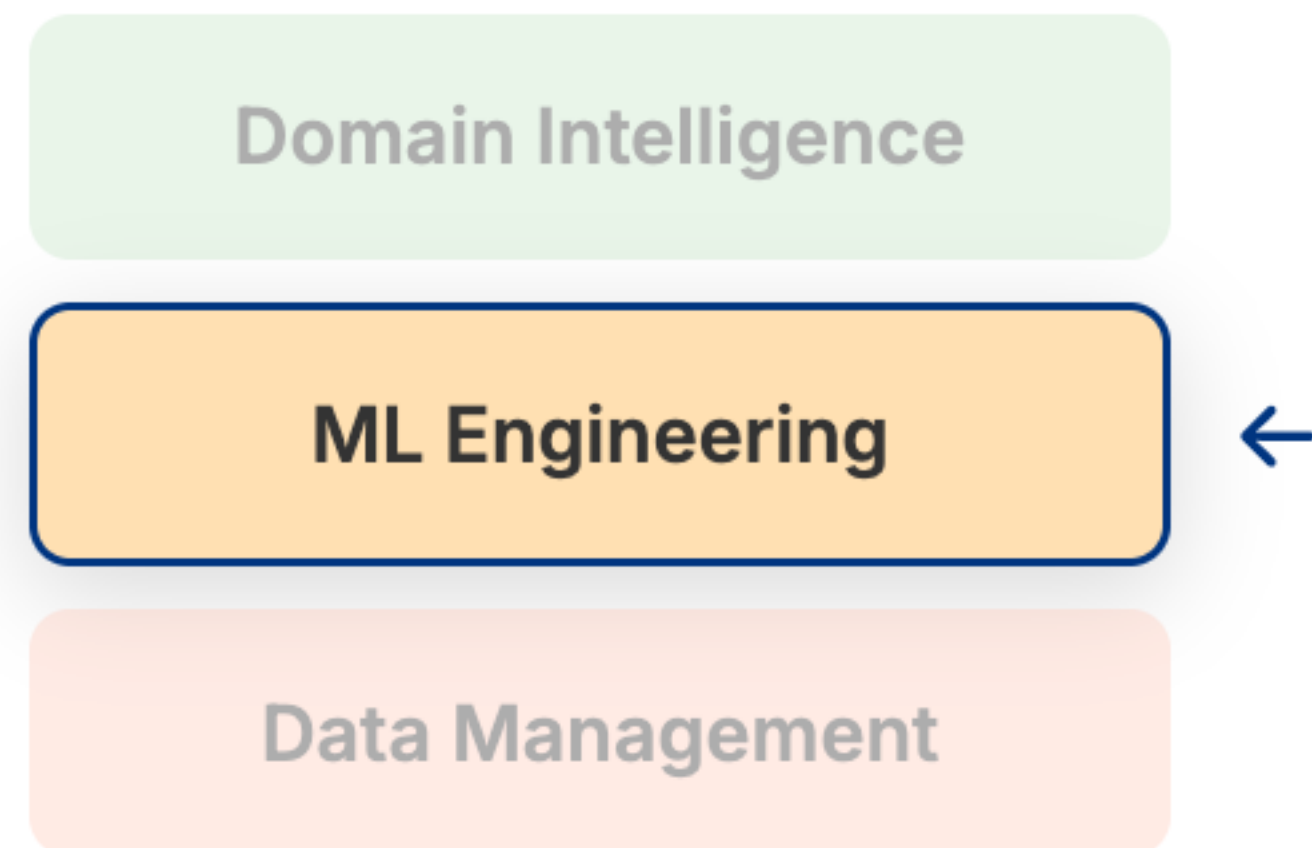


Build Quality Gates



Track Lineage End-to-End

ML Engineering Layer



ML Model Serving + Feature Engineering

Infrastructure powering AI at scale

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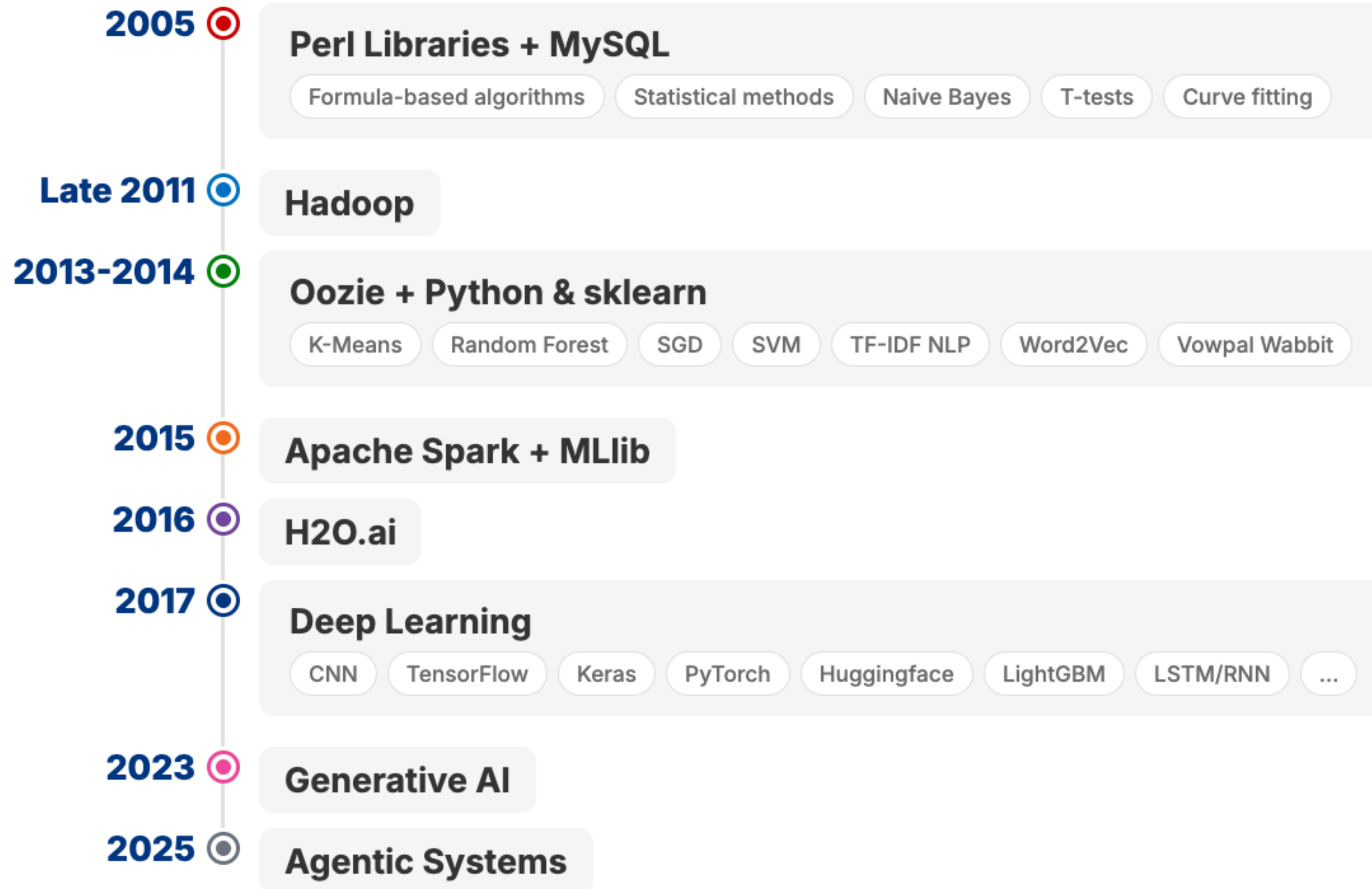
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Evolution of our ML Stack



2015 was A Pivotal Year

We began solving two core problems



1. Realtime Predictions

Online Inference at scale

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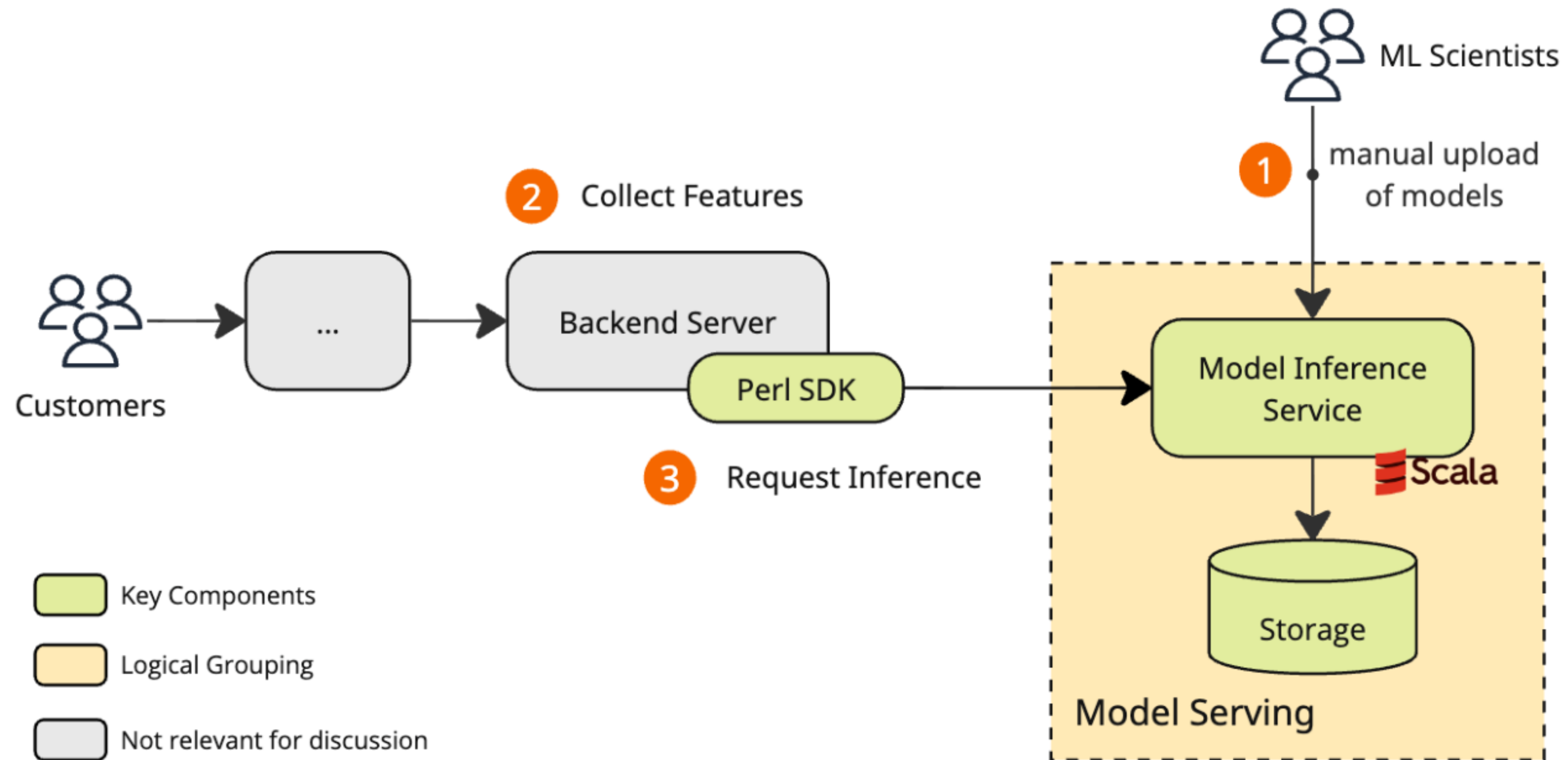
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Our First Model Serving Service

Abstracting inference complexity



Supported Model Formats



H2O Models



TensorFlow 2.x



Python Models



Vowpal Wabbit



Ensemble (Chimera)



LightGBM



SageMaker (AWS)



Triton (NVIDIA)



Linear Models (CSV)



Lookup Tables (Cassandra)



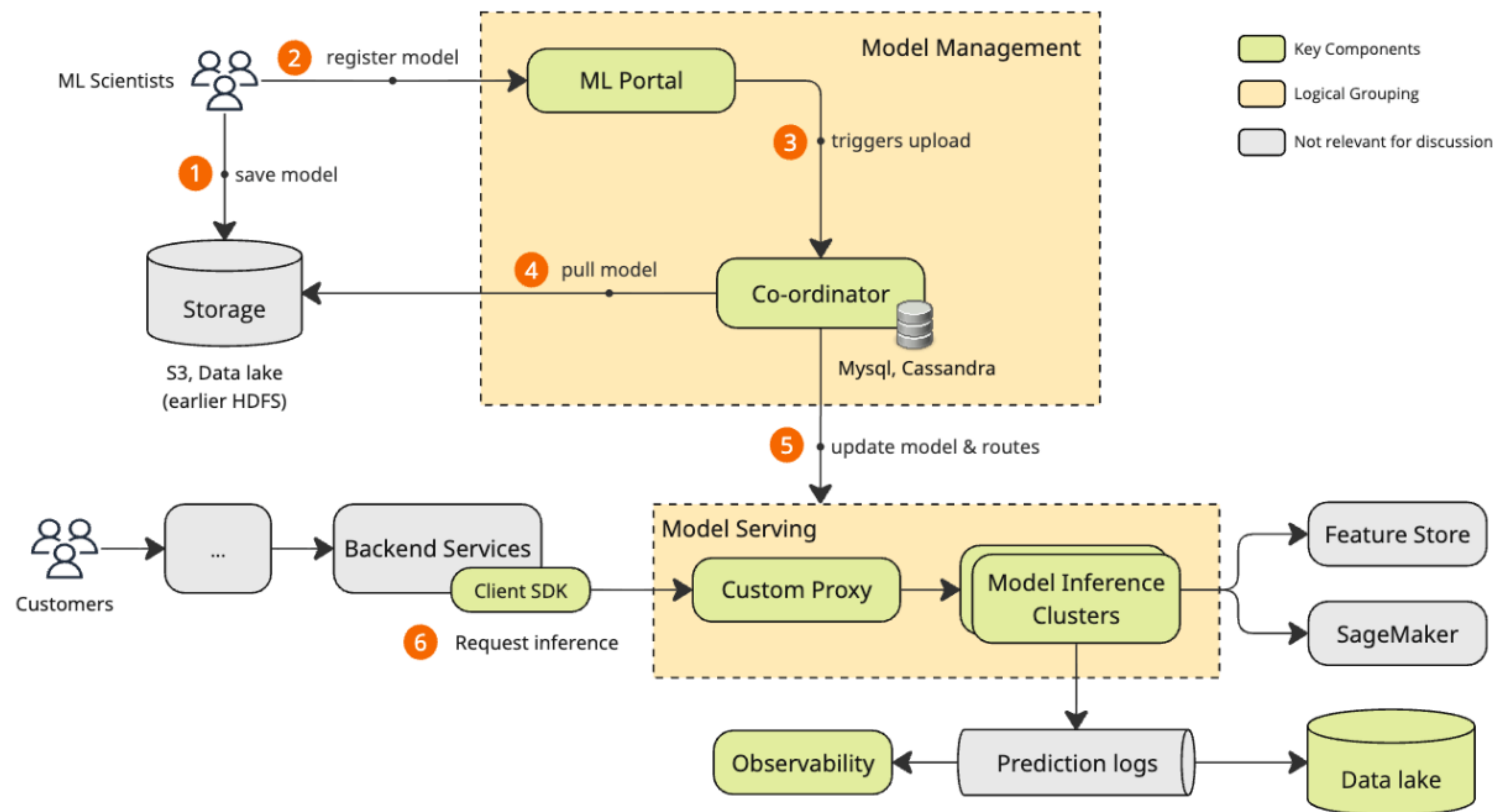
Identity Models



Legacy Models

Today's ML Inference Platform

Simplified Inference Architecture



Numbers as of 2024

480+

ML Models

400B+

Predictions Per Day

< 20ms

Typical Latency

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2015 was A Pivotal Year

We began solving two core problems



1. Realtime Predictions

Online Inference at scale



2. Feature Engineering

For Training & Inference

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ML Scientists' Spent **80% of Their Time** in Munging Data

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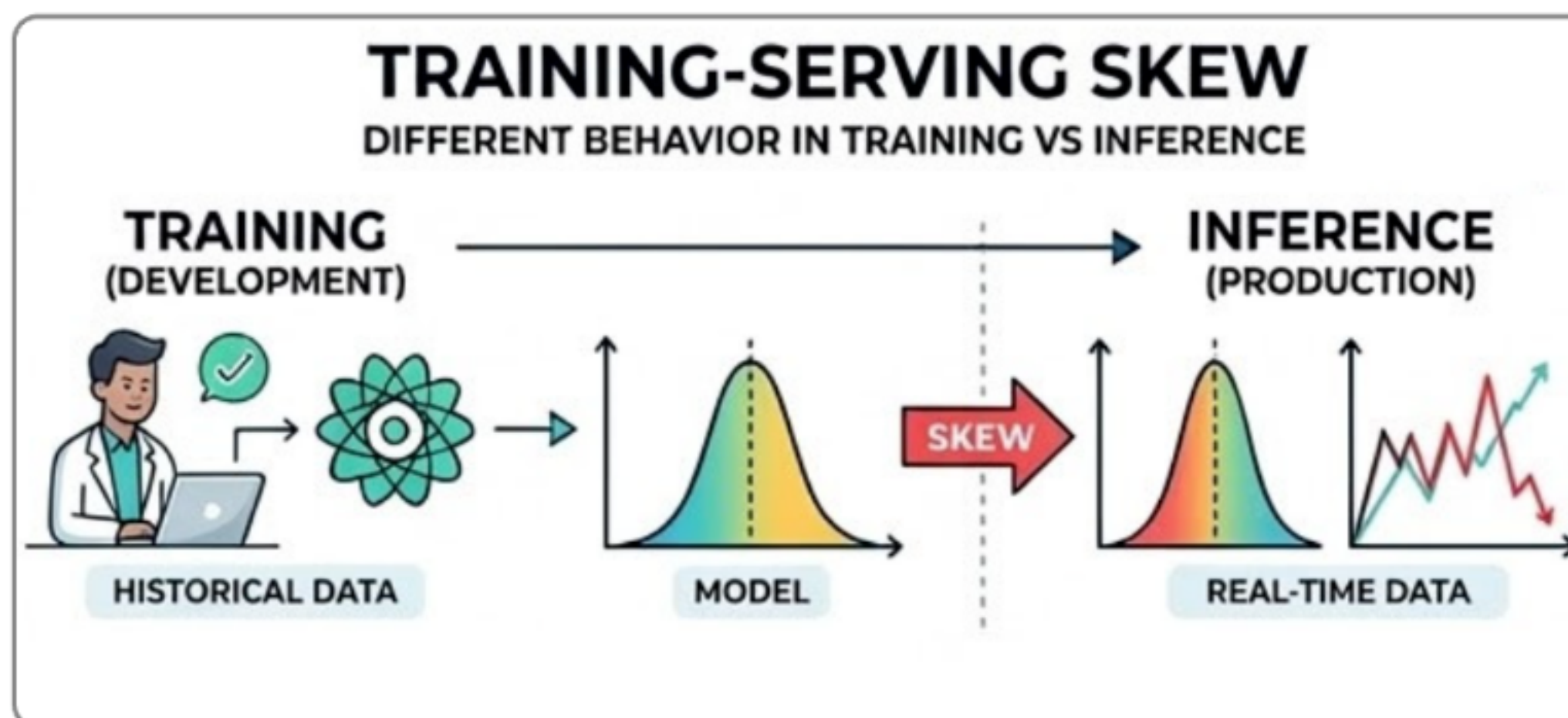
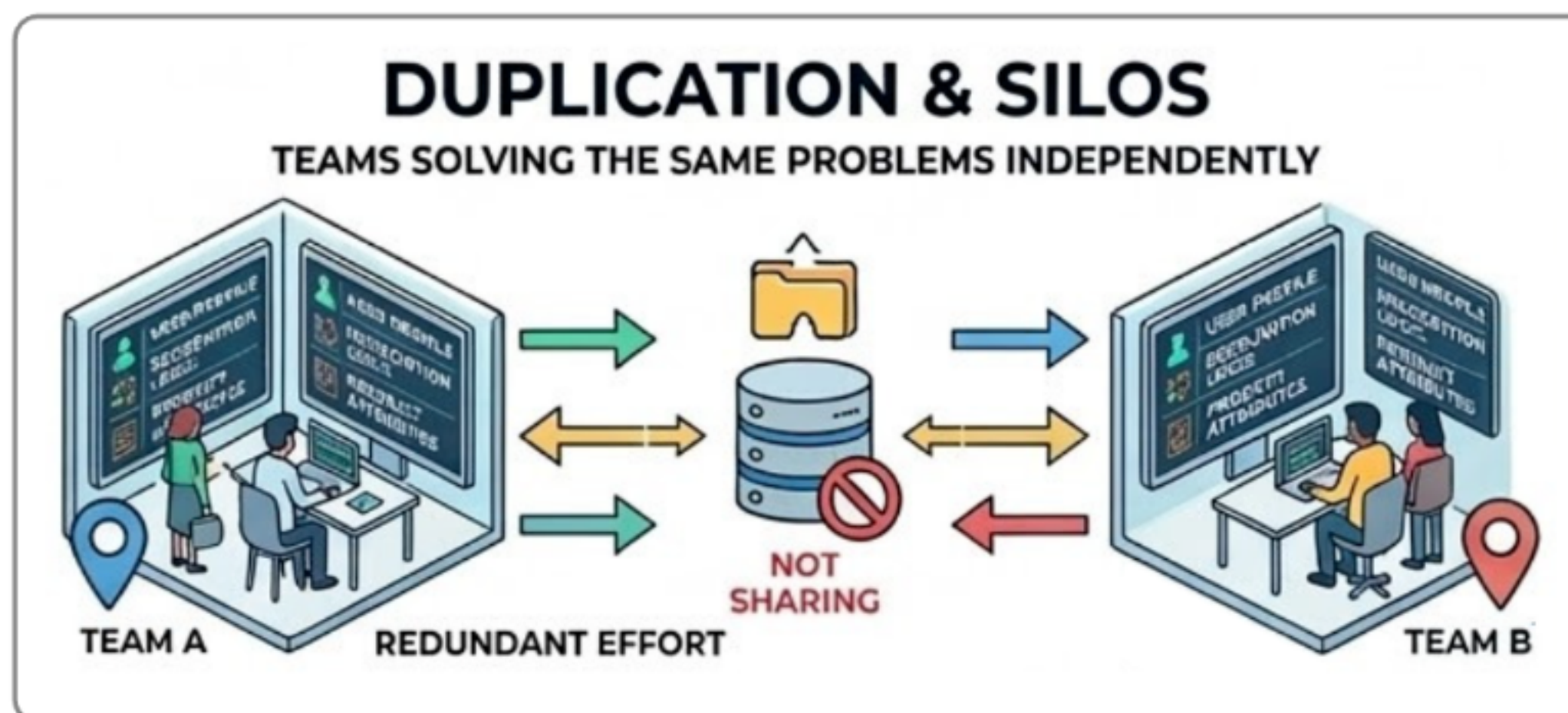
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Two Pain Points Emerged



Let's Build **Re-usable** Computations

Our First Attempt

Offline Computation

For Training Features

Online Computation

For Inference Features

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Building Training Dataset

A Classic Approach

user_id	search_timestamp	destination	did_book
abc123	2016-08-06 12:00:00	paris	0
abc123	2016-08-07 15:30:00	amsterdam	1
xyz789	2016-08-06 14:00:00	madrid	0

Initial Dataset



user_id	timestamp	search_count_17d
abc123	2016-08-05 10:00:00	1
abc123	2016-08-05 14:30:00	2
abc123	2016-08-06 09:15:00	3
abc123	2016-08-06 16:45:00	4
abc123	2016-08-07 11:20:00	5
xyz789	2016-08-05 08:00:00	1
xyz789	2016-08-06 10:30:00	2

Feature Table



user_id	search_timestamp	destination	did_book	search_count_17d
abc123	2016-08-06 12:00:00	paris	0	3
abc123	2016-08-07 15:30:00	amsterdam	1	5
xyz789	2016-08-06 14:00:00	madrid	0	2

Training Dataset

Let's Help Build Feature Tables and Time-wise Joins

in a consistent manner

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How did We Help Build Feature Tables?

Example: Count the no. of searches in the last 7 days

user_id	timestamp	destination	...
abc123	2016-08-05 10:00:00	paris	...
abc123	2016-08-05 14:30:00	london	...
abc123	2016-08-06 09:15:00	amsterdam	...
abc123	2016-08-06 16:45:00	berlin	...
abc123	2016-08-07 11:20:00	rome	...
xyz789	2016-08-05 08:00:00	madrid	...
xyz789	2016-08-06 10:30:00	barcelona	...

Fact Log

Eg., Log of Search Events



Converts one DataFrame to another

```
/**
 * Count how many searches occurred in the 7-day window before current timestamp.
 */
class SearchCountL7dTransformer(override val uid: String) extends Transformer {
  def this() = this(Identifiable.randomUUID("search_count_l7d"))
  override def transform(dataset: Dataset[_]): DataFrame = {
    ...
  }
}
```

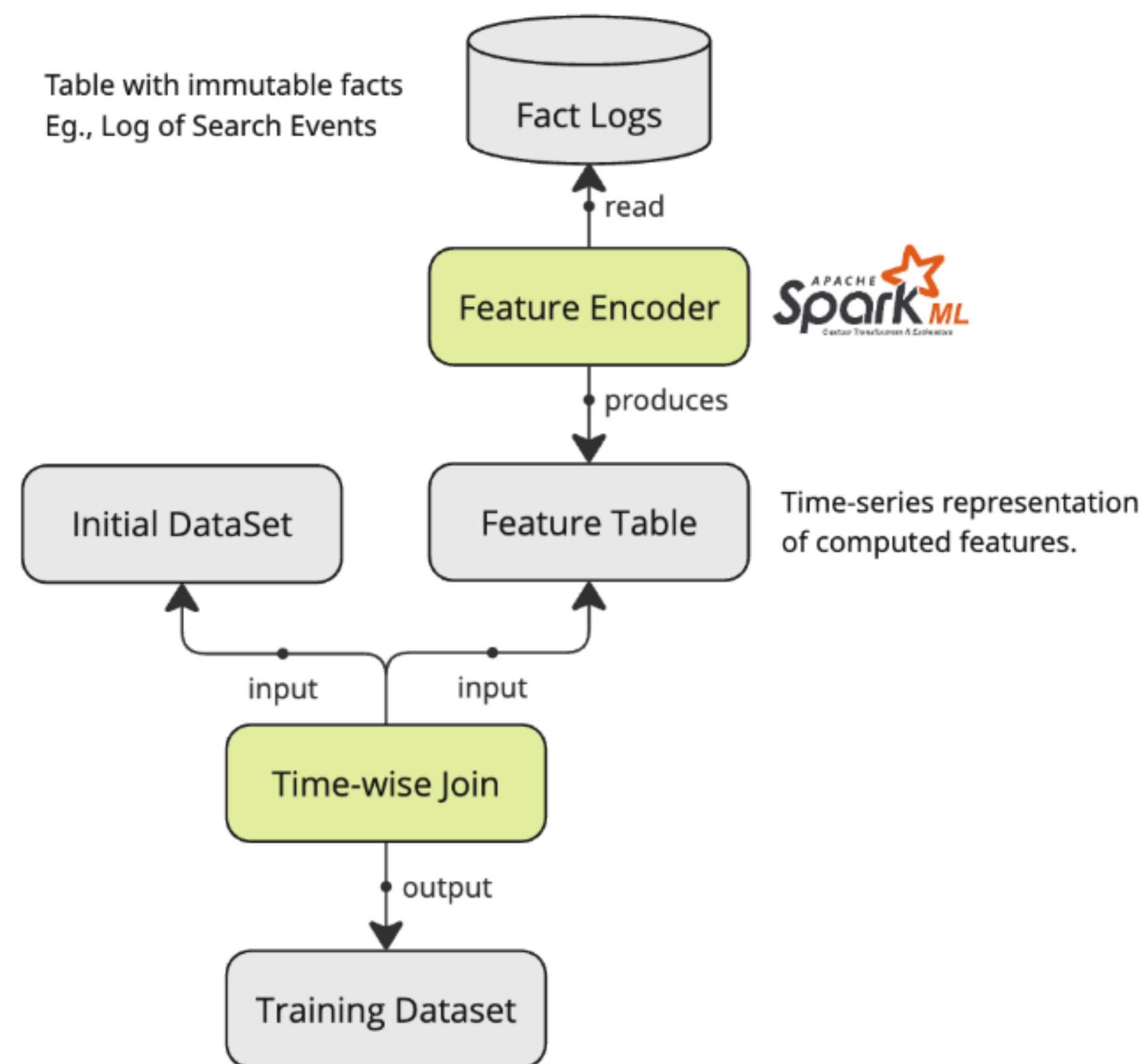
Feature Encoder

user_id	timestamp	search_count_l7d
abc123	2016-08-05 10:00:00	1
abc123	2016-08-05 14:30:00	2
abc123	2016-08-06 09:15:00	3
abc123	2016-08-06 16:45:00	4
abc123	2016-08-07 11:20:00	5
xyz789	2016-08-05 08:00:00	1
xyz789	2016-08-06 10:30:00	2

Feature Table

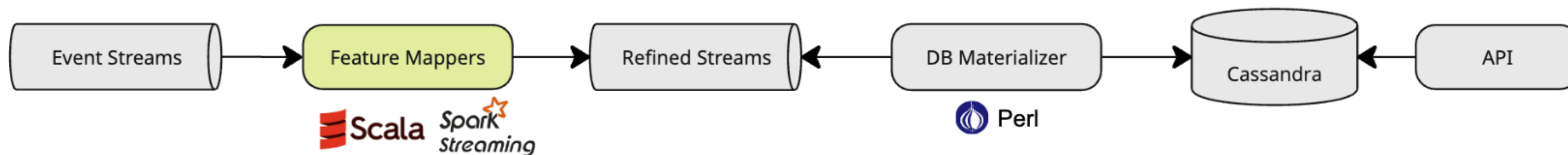
Offline Feature Computation

Using SparkML Transformers



Online Feature Computation

Computing features in real-time



But There Were Challenges

Parity • Skew • Performance • Stability • Backfilling

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Let's Bring **Data & Computational Parity**

Our Second Attempt

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We Tackled **Data Parity** with **Timelines**

Time-ordered sequence of events per user

Example: Search logs, Reservations, Cancellations

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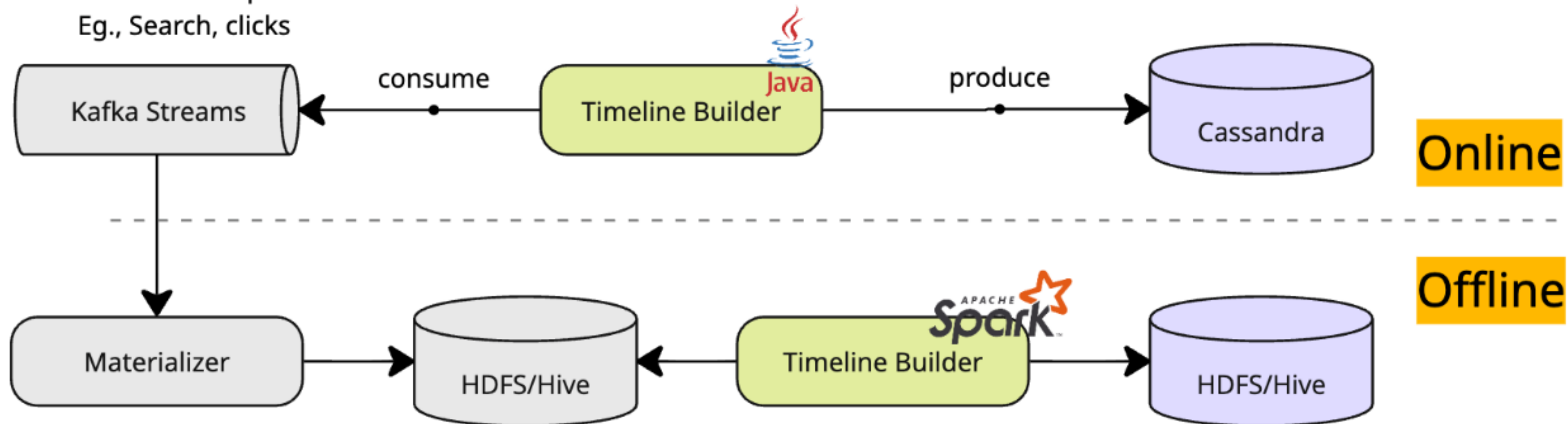
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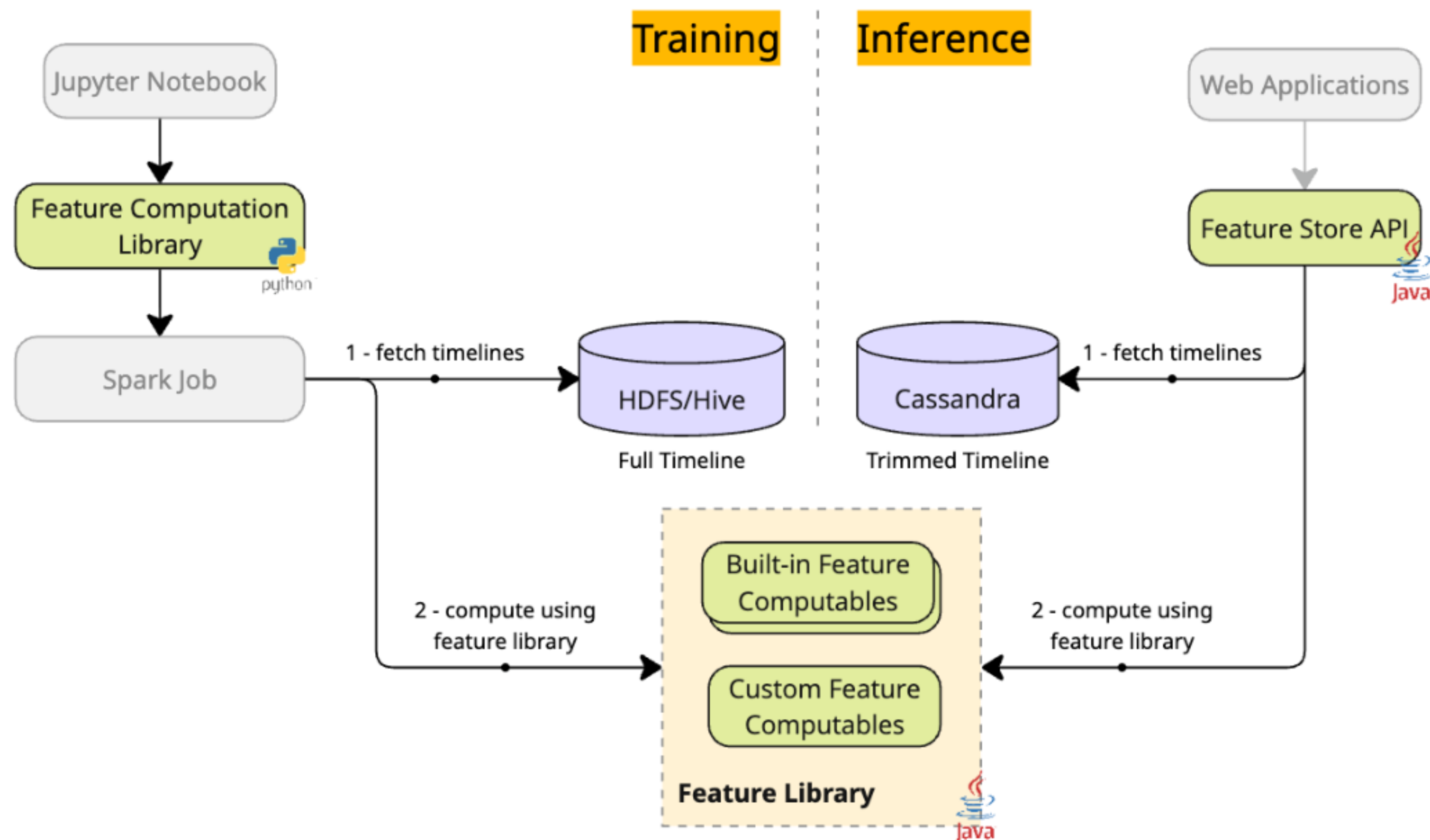
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How Were Timelines Built?

Immutable time-stamped actions
Eg., Search, clicks



How We Used Timelines for Computational Parity?



But There Were Major Downsides

Java Requirement • High Latency

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Decentralize **Feature Computation** and Centralize **Feature Storage and Serving**

Our Third Attempt

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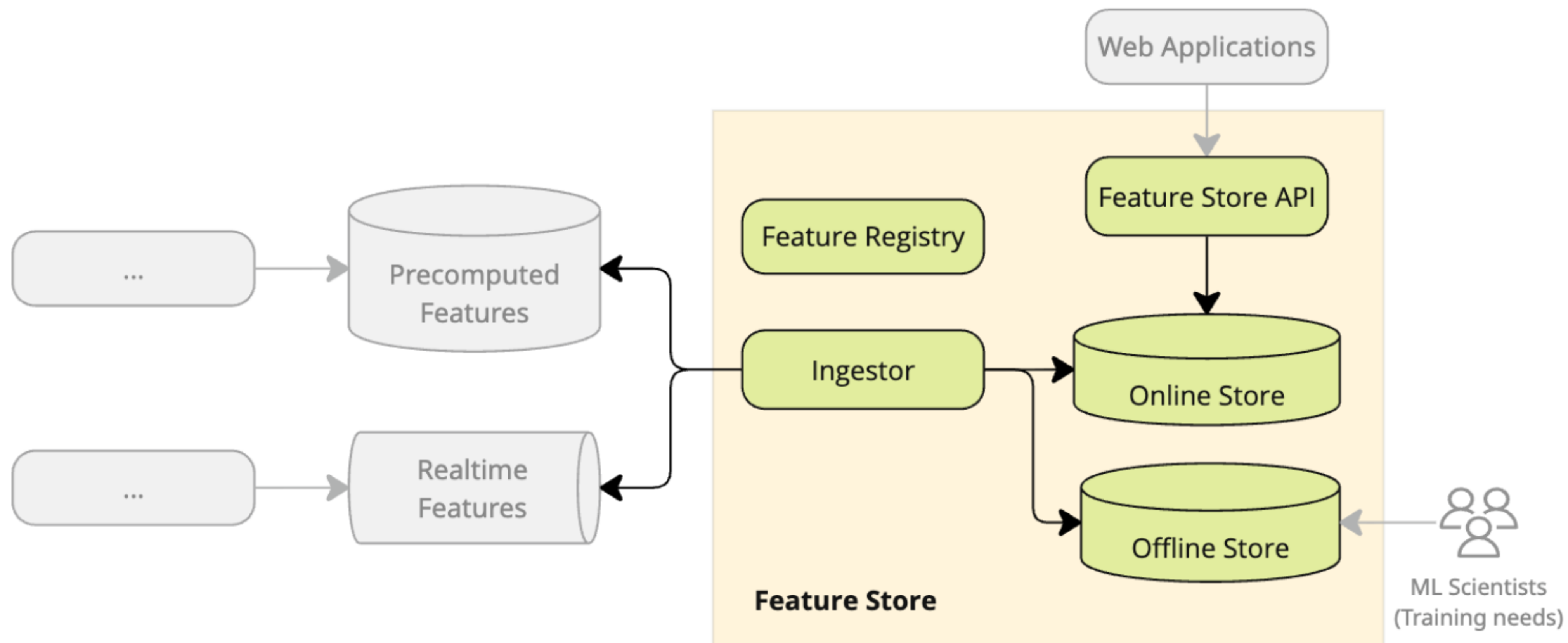
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Feature Store

Manage, store, and serve feature data





Feature Engineering Is a Multi-Dimensional Problem

And we haven't found a silver bullet yet!

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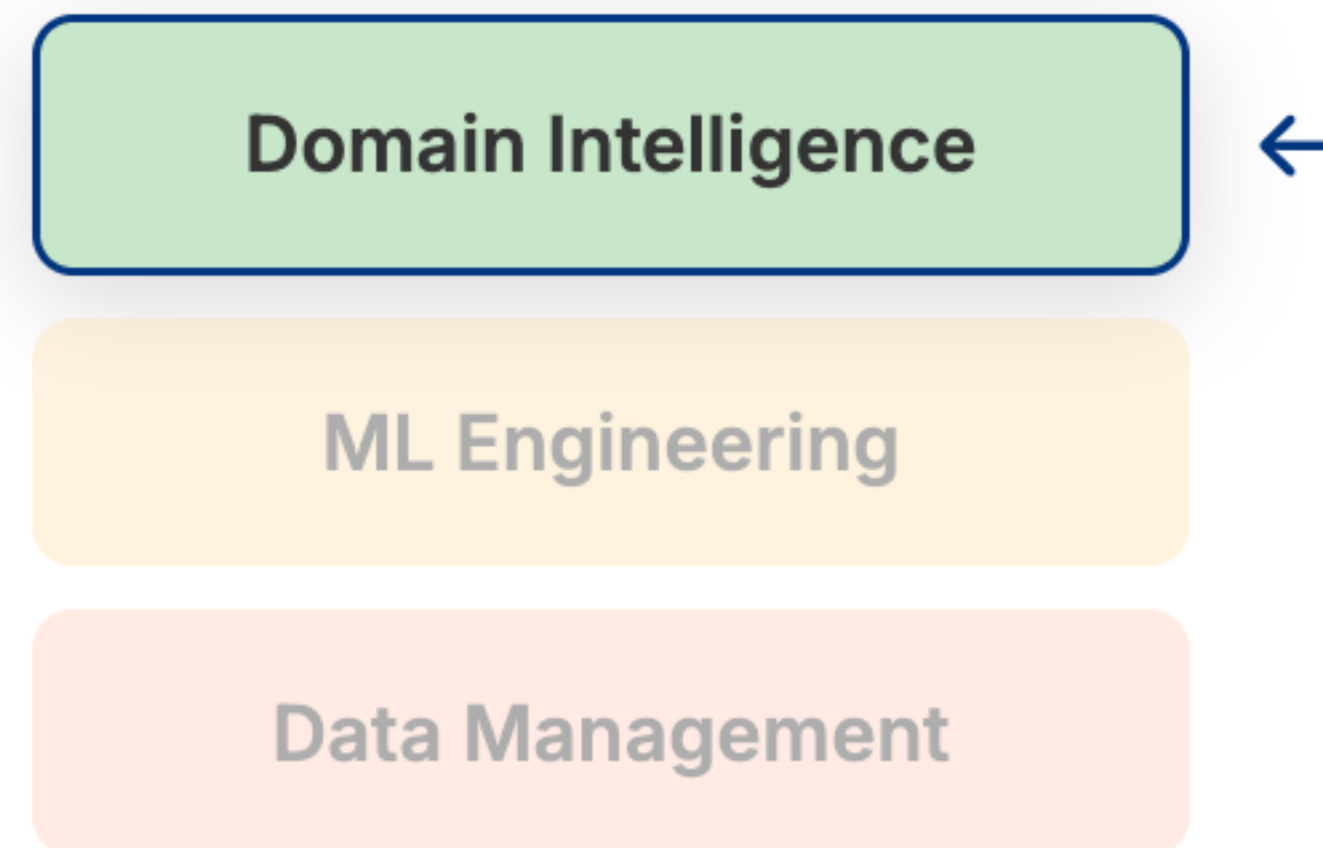
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Domain Intelligence Layer



A Collection of Domain-Specific ML Platforms

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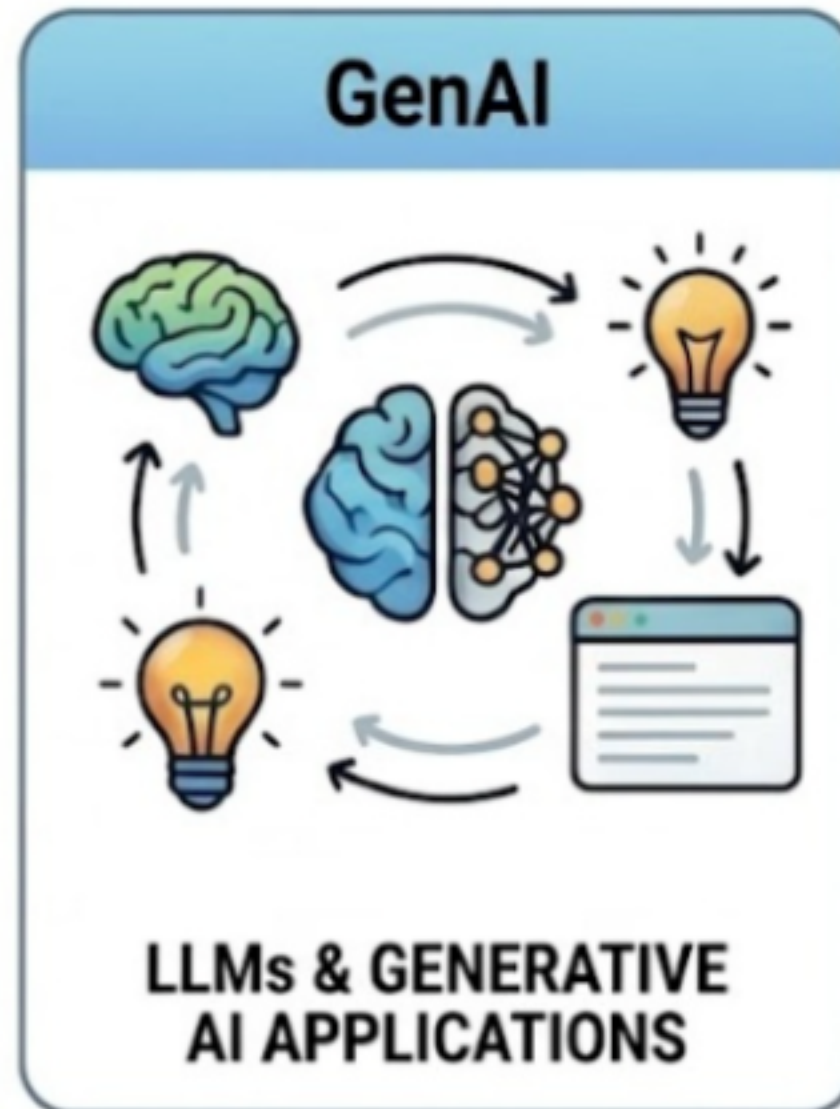
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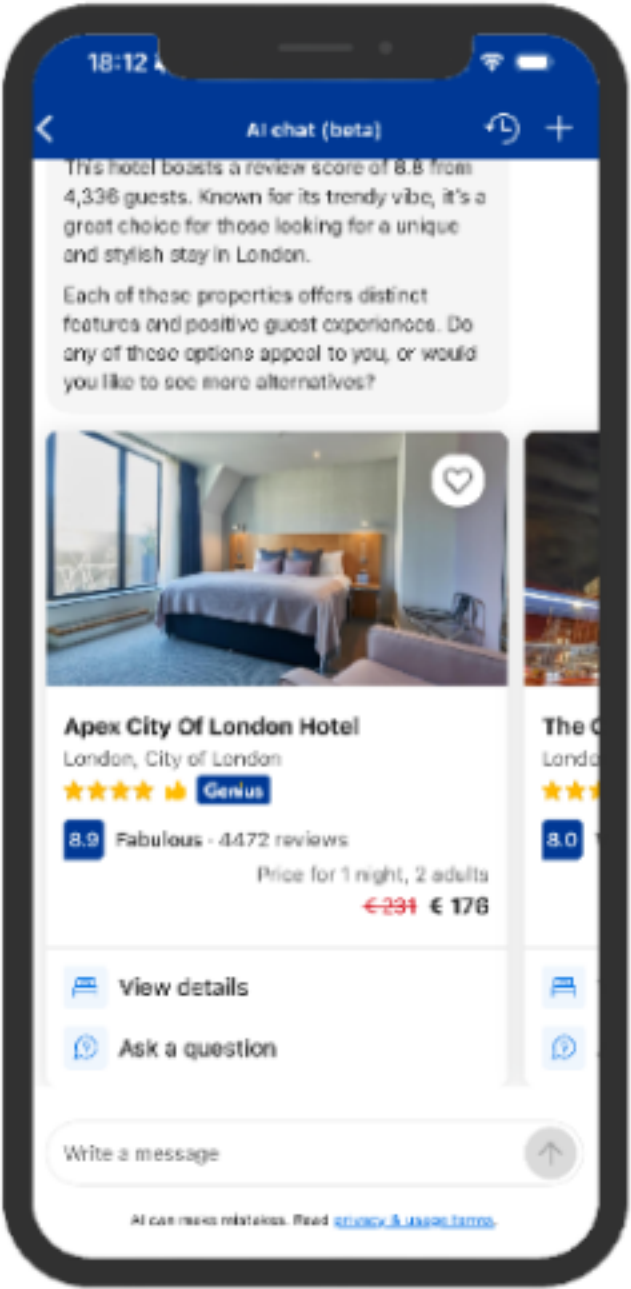
SUMMARY

Domain-Specific ML Platforms

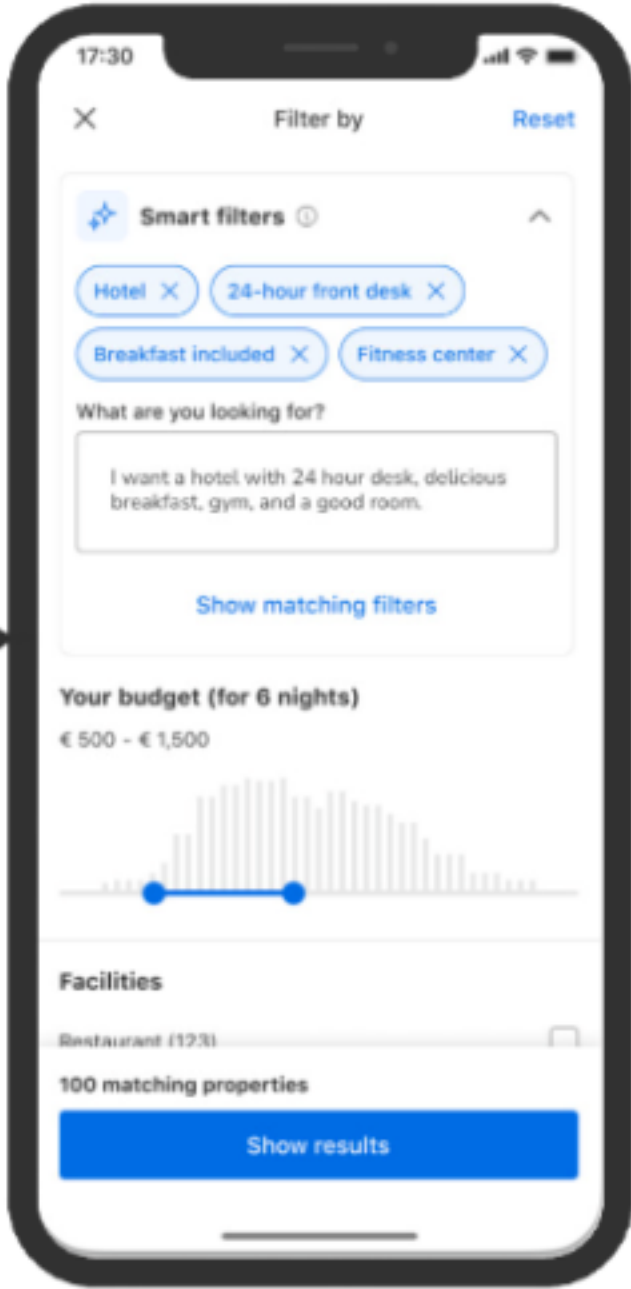
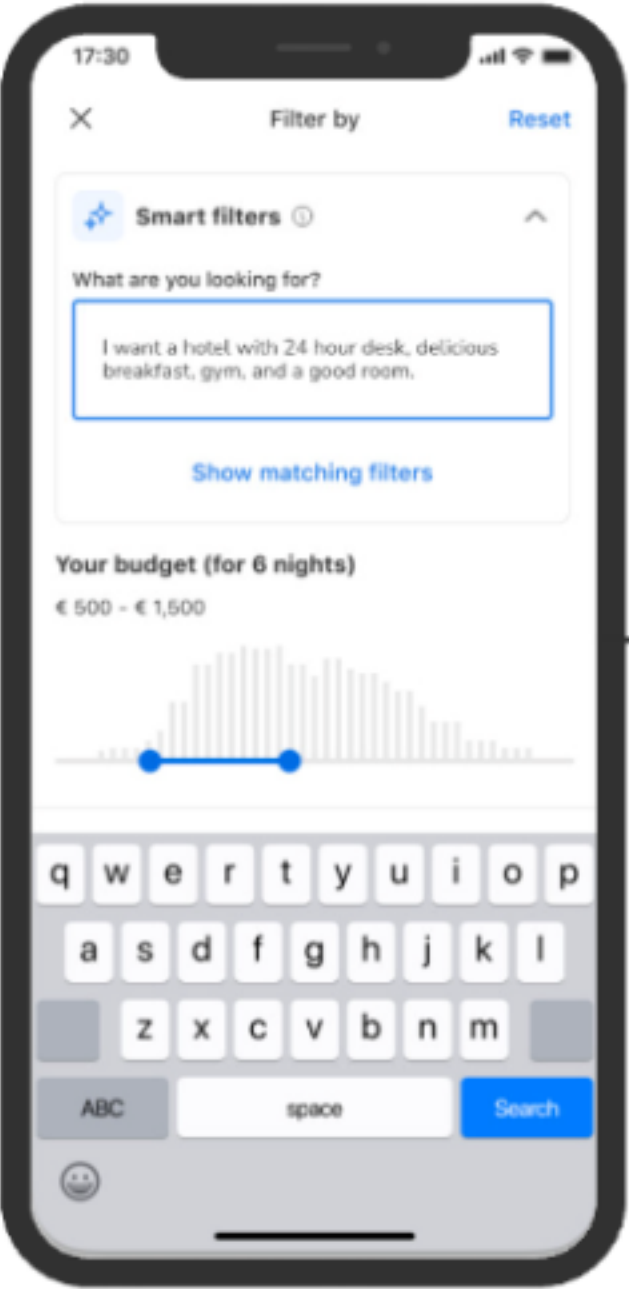


GenAI Usecases

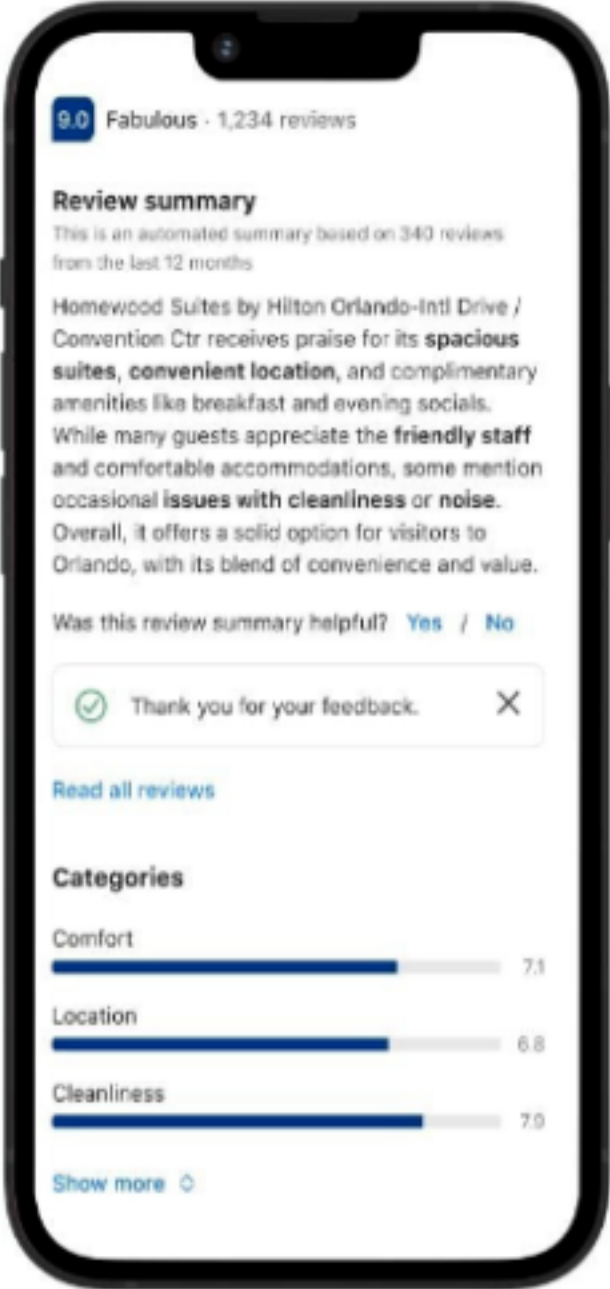
Trip Planning



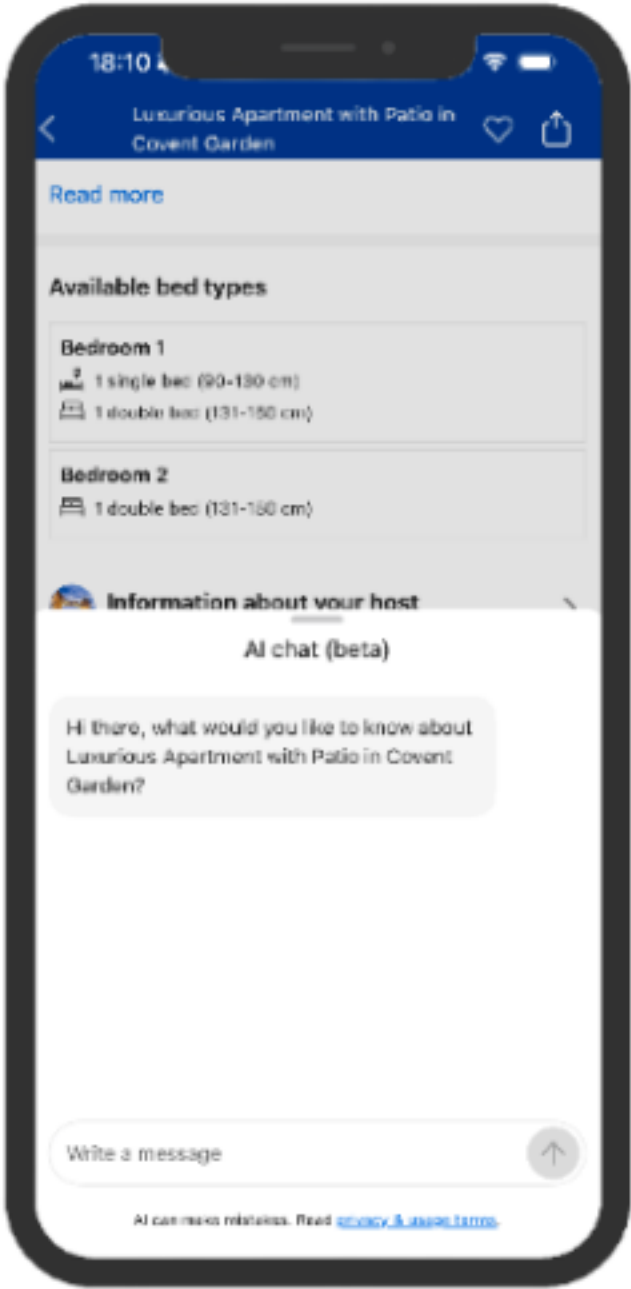
Smart Filters



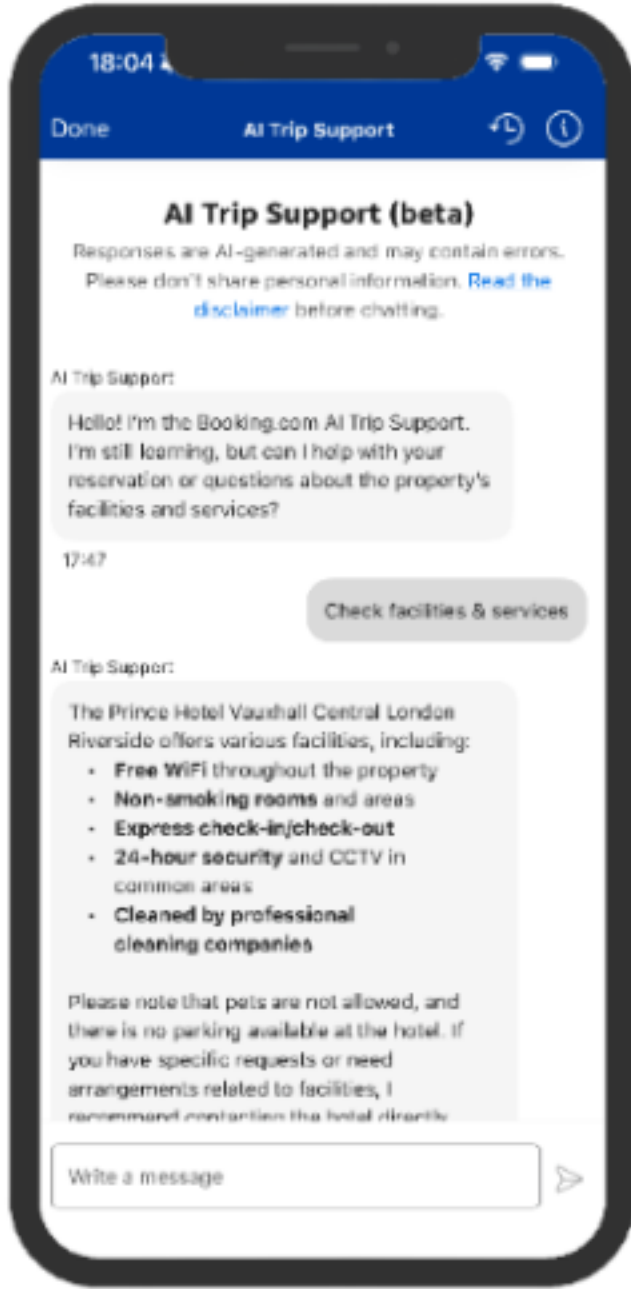
Review Summaries



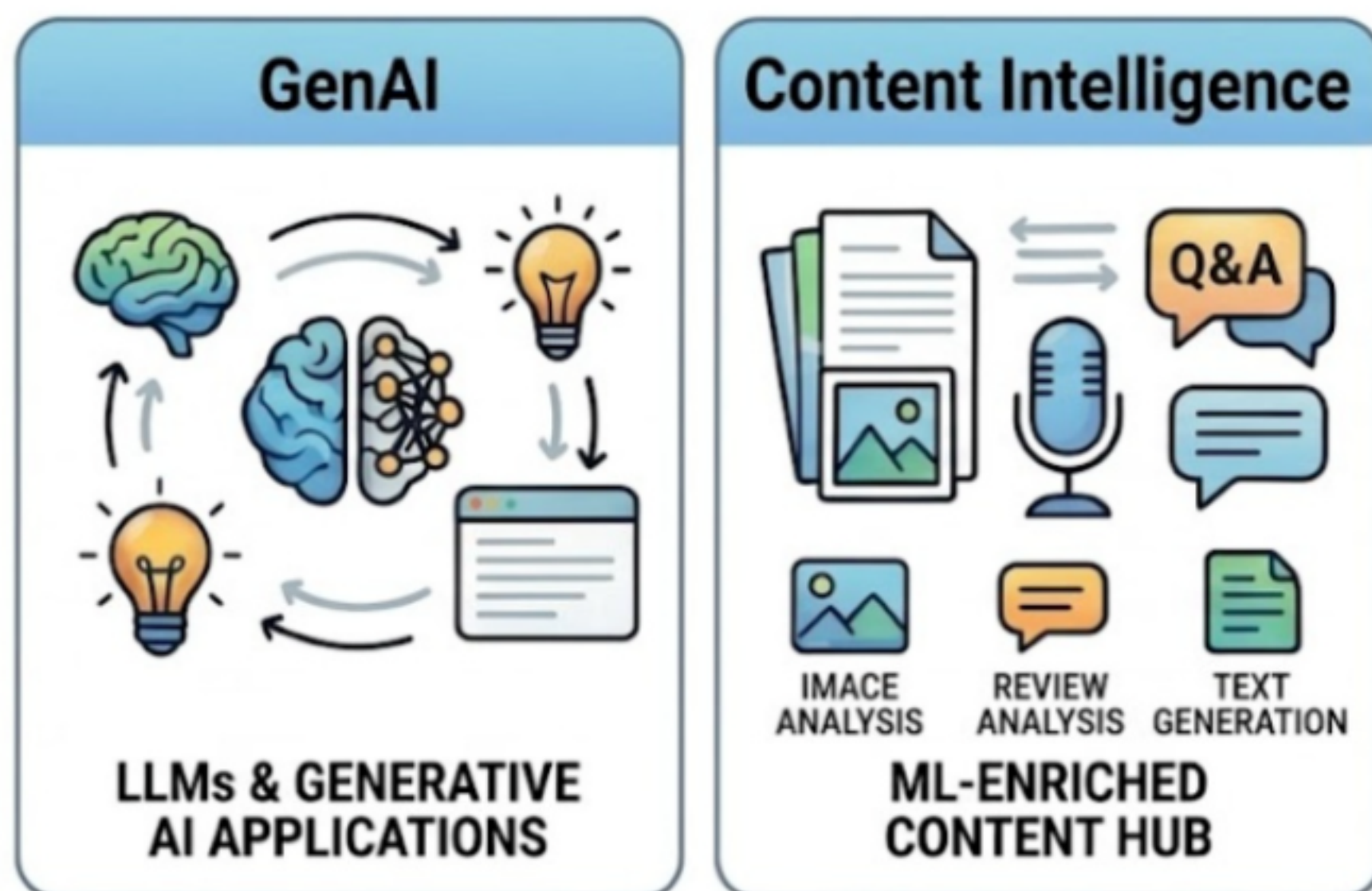
Property QnA



Post-Booking Support



Domain-Specific ML Platforms



Content Intelligence Usecases

Overview Apartment Info & Price Facilities House rules Important and legal info Guest reviews (3)

Family-Friendly Retreat Westfield Holland Park

St. Stephens Avenue, Hammersmith and Fulham, London, W12 8JH, United Kingdom – [Good location – show map](#) – [Subway Access](#)

[Reserve your apartment stay](#)

[We Price Match](#)

Excellent 8.7
3 reviews

Guests who stayed here loved
"Interesting layout, authentic house, lots of space."

Staff 9.2
Violeta Lithuania

2 BEDROOMS **FREE WIFI** **LIVING ROOM** **KITCHEN**

+27 photos

The entire place is yours **80 m² size** **Private bathroom** **Free Wifi** **Bathtub**

Kitchen **Washing machine** **Cable channels** **Heating**

[Show on map](#)

Guest review summary
Automated summary based on 622 reviews from the past 12 months.

- Location:** Central and quiet setting near Dam Square, shops, and canals.
- Staff:** Friendly, professional team provide a warm and helpful welcome.
- Breakfast:** Buffet offers a wide variety of fresh pastries, fruit, and hot dishes.
- Cleanliness:** Housekeeping keeps spaces spotless and immaculate.

Room Size: Upgraded units are spacious, but standard options feel small and tight.

Was this guest review summary helpful? [Yes](#) / [No](#)

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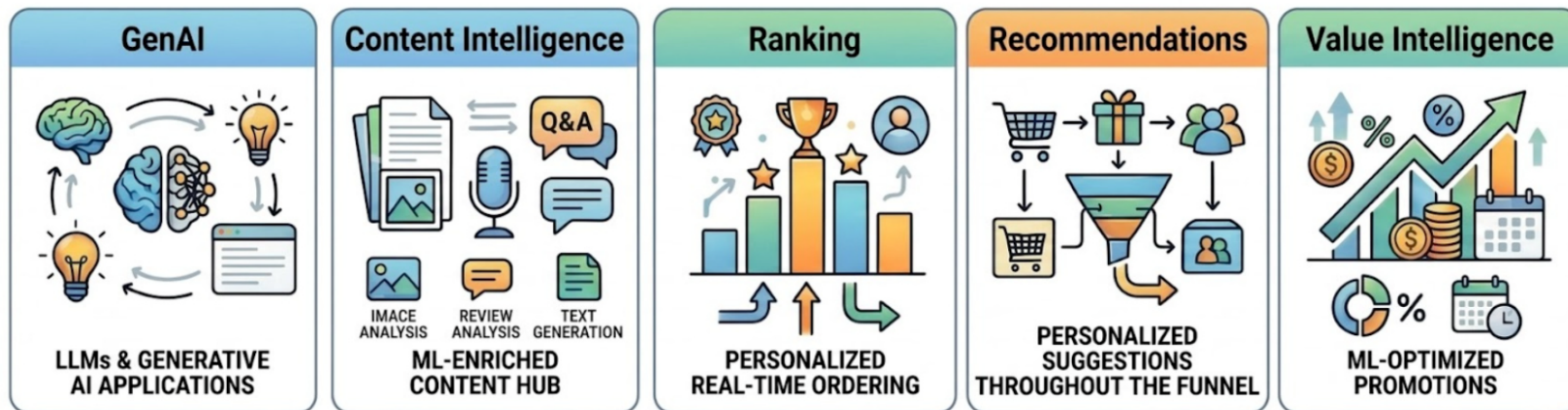

Domain-Specific ML Platforms



Recommendations Usecases



Domain-Specific ML Platforms



Ranking Platform

Personalized real-time ordering at scale

The Three-way Optimization Challenge

Traveller • Partner • Business



INTRO

2005

EXPERIMENTATION

DATA

MYSQL

HADOOP

CLOUD

ML ENGINEERING

ML INFERENCE

FEATURE ENGINEERING

DOMAIN INTELLIGENCE

RANKING

SUMMARY

Ranking, Circa 2005

A simple formula

```
Hotel Rank = F( Bookings, views, ... ) + rand()
```


The Formula Evolved

Some factors that were experimented

CANCELLATIONS



VISITOR TRAFFIC VOLUME



DISTANCE BASED RANKING



FACTORING IN ROOM AVAILABILITY



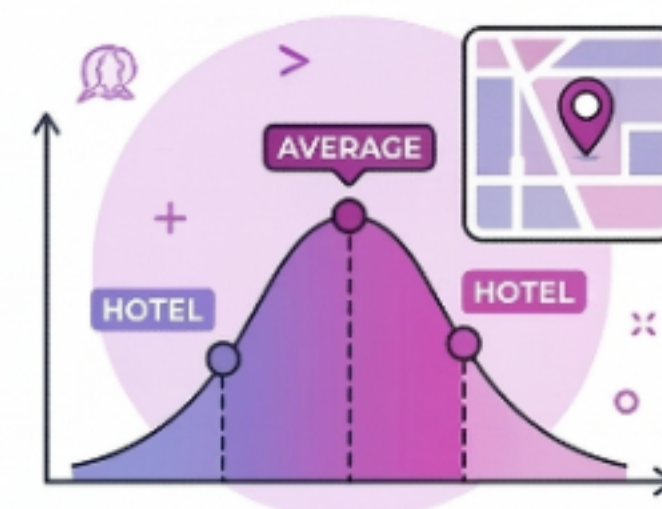
BASED ON GUEST REVIEWS



SEASONALITY AWARENESS



AVERAGE RANKING IN THE LOCATION



USING HOTEL IMPRESSIONS



We Attempted to **Replace The Formula with ML**

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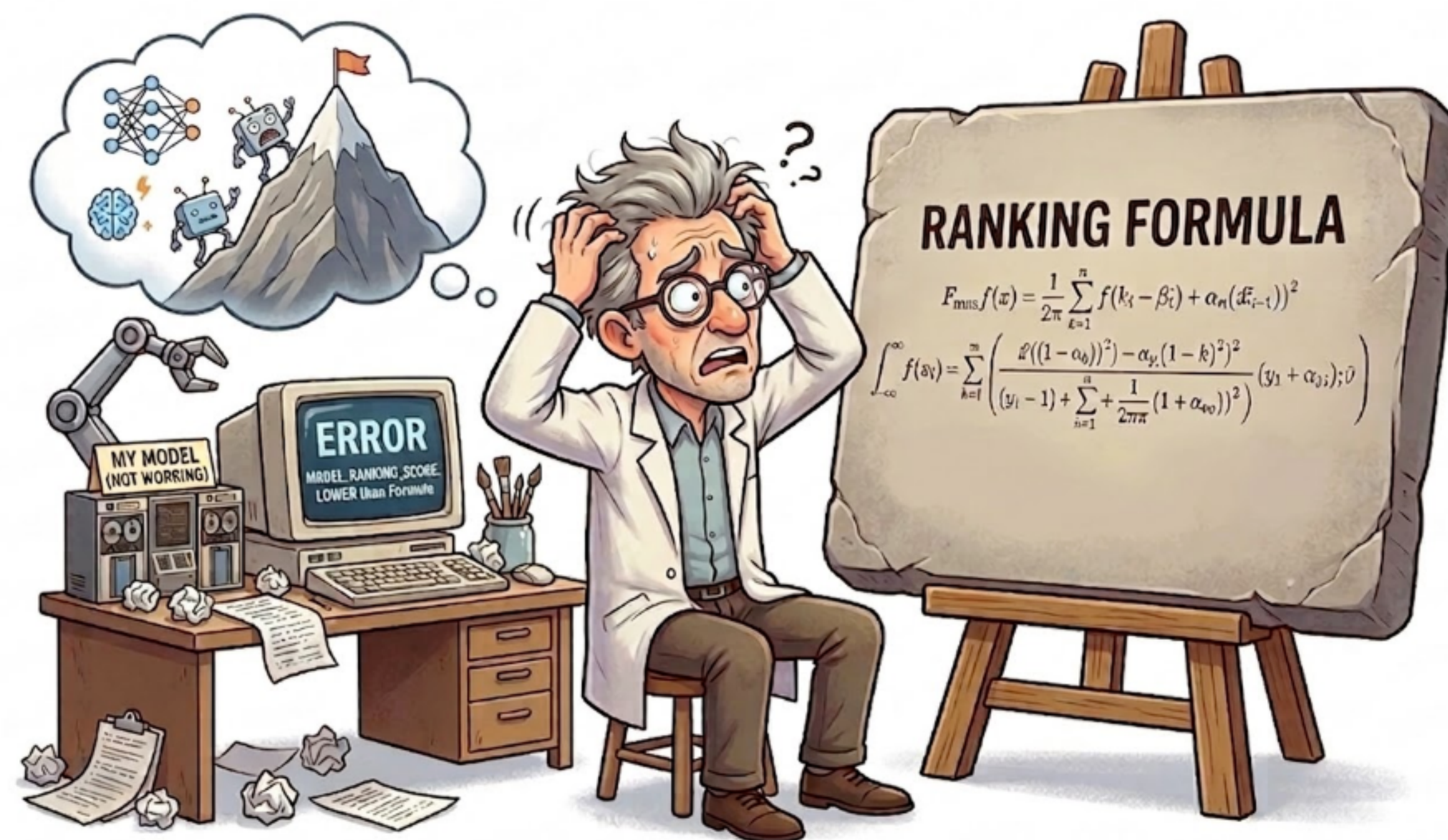
DOMAIN INTELLIGENCE

RANKING

SUMMARY

But **The Hand-crafted Formula** was Undefeatable ...

... for a longtime



Chief Reason was **Infrastructure Limitation**



Next, We Wanted to Tackle **Personalized Ranking**

But not many believed it was possible

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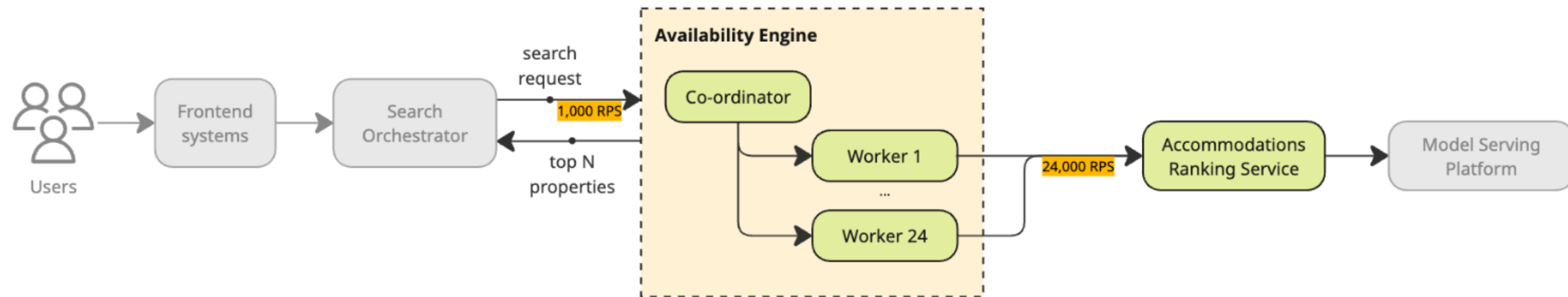
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DOMAIN INTELLIGENCE

RANKING

SUMMARY

The Challenge with Realtime Personalization



High-throughput & Low-latency requirements

Throughput and Latency

Numbers from 2024

100K+

requests/second

<100ms

p99.9 latency

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Experiments Typically Run for **2-4 weeks**

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But, We Wanted to **Move Faster**

Well aware that poor ranking models are expensive.

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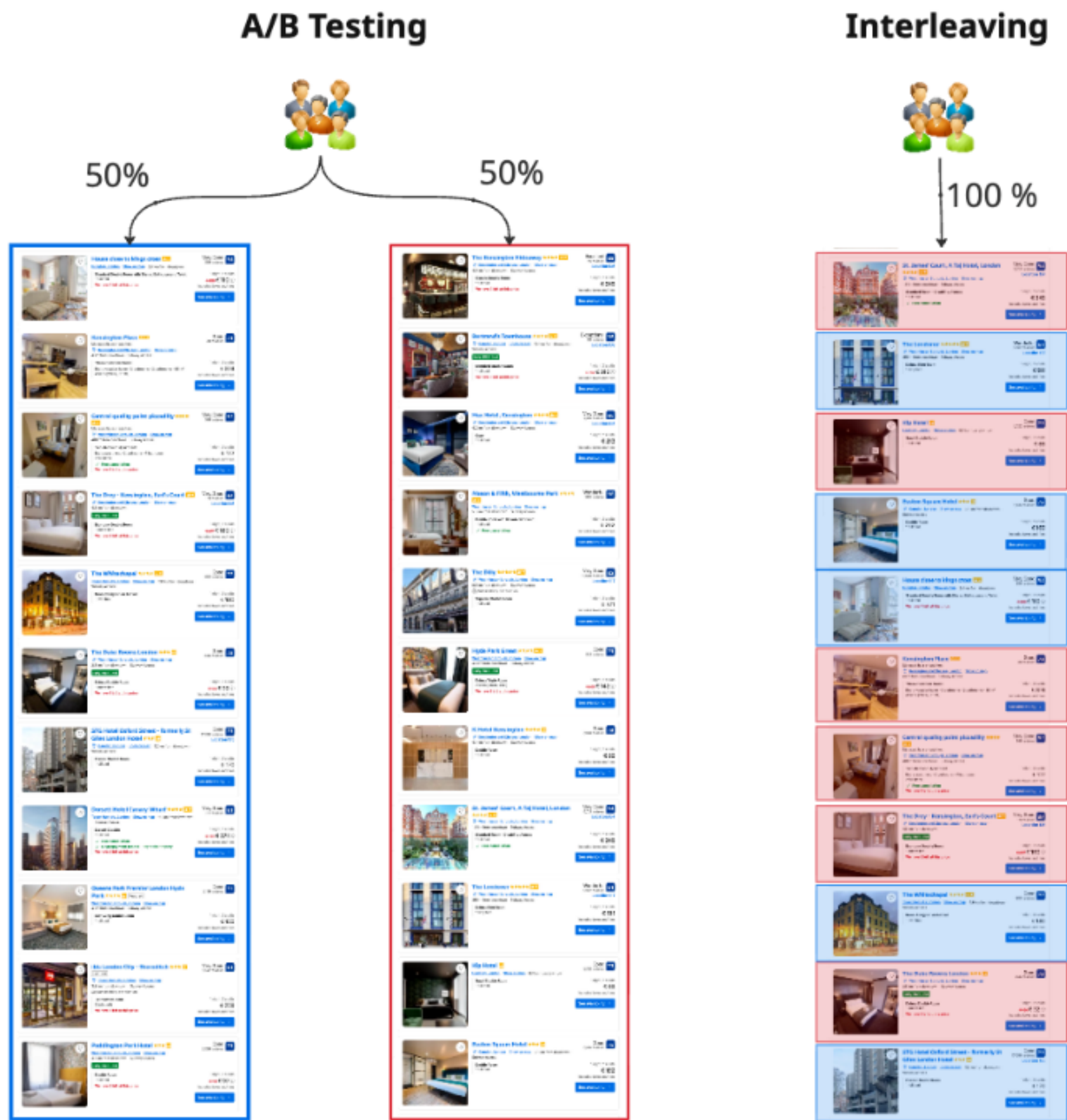
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SUMMARY

Interleaving Unlocked Faster Experimentation



Key Interleaving Benefits

More Variants Together

10x Speedup

On Less Traffic

100x less

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Preselect with **Interleaving** & Validate with **A/B Testing**

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Experimentation still underpins everything we do

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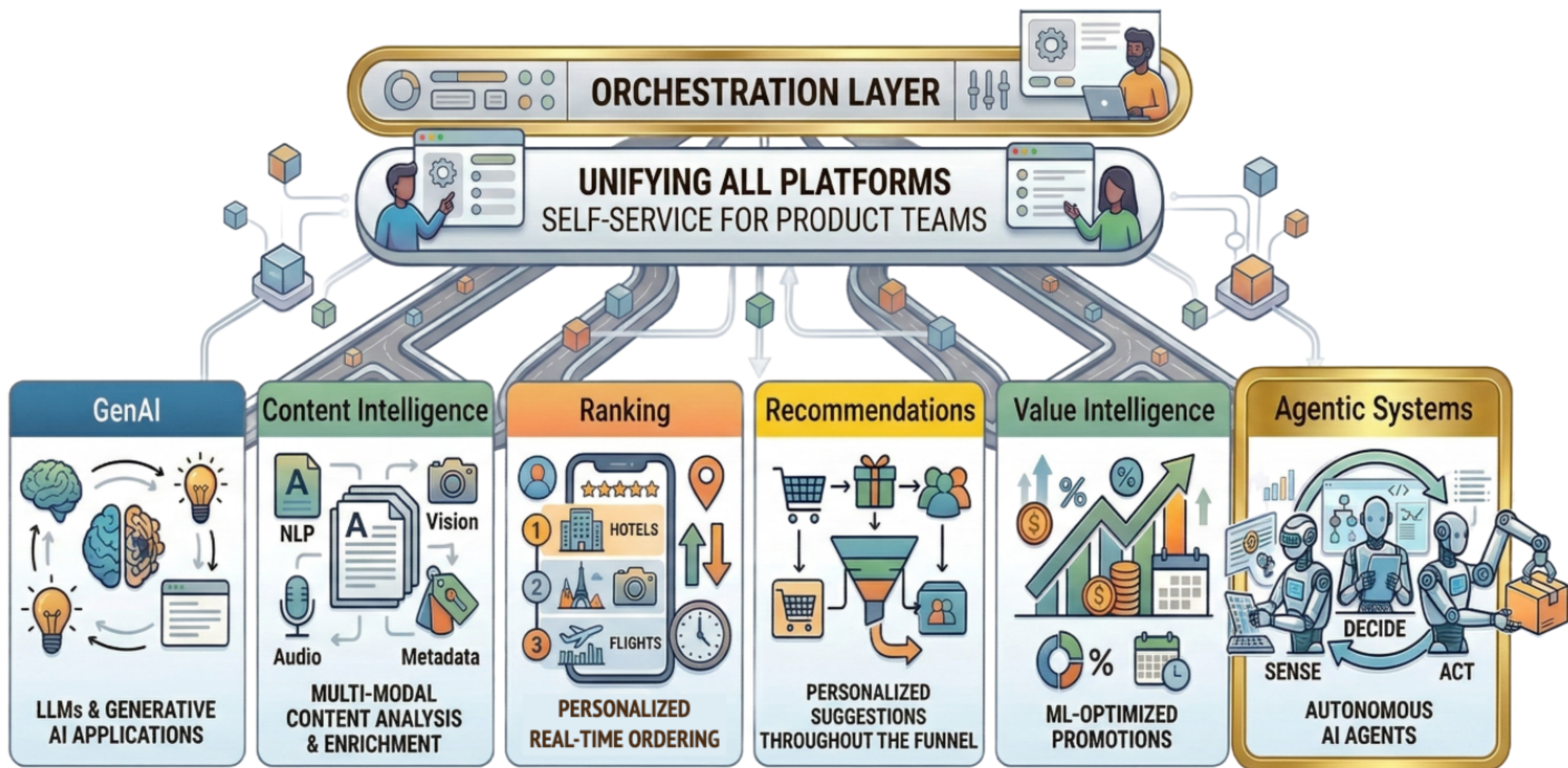
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SUMMARY

What's Brewing?





Jabez Eliezer Manuel

Principal Engineer @ Booking.com



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